

NAVRAP · AIRNAV RISK ANALYSIS PLATFORM

User Guide

How to plan, run, review and approve safety risk assessments
with the twelve methods of the AirNav Risk Analysis Manual

Software version 0.9 · Guide edition 9 · September 2026

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NAVRAP User Guide

AirNav Risk Analysis Platform — version 0.9

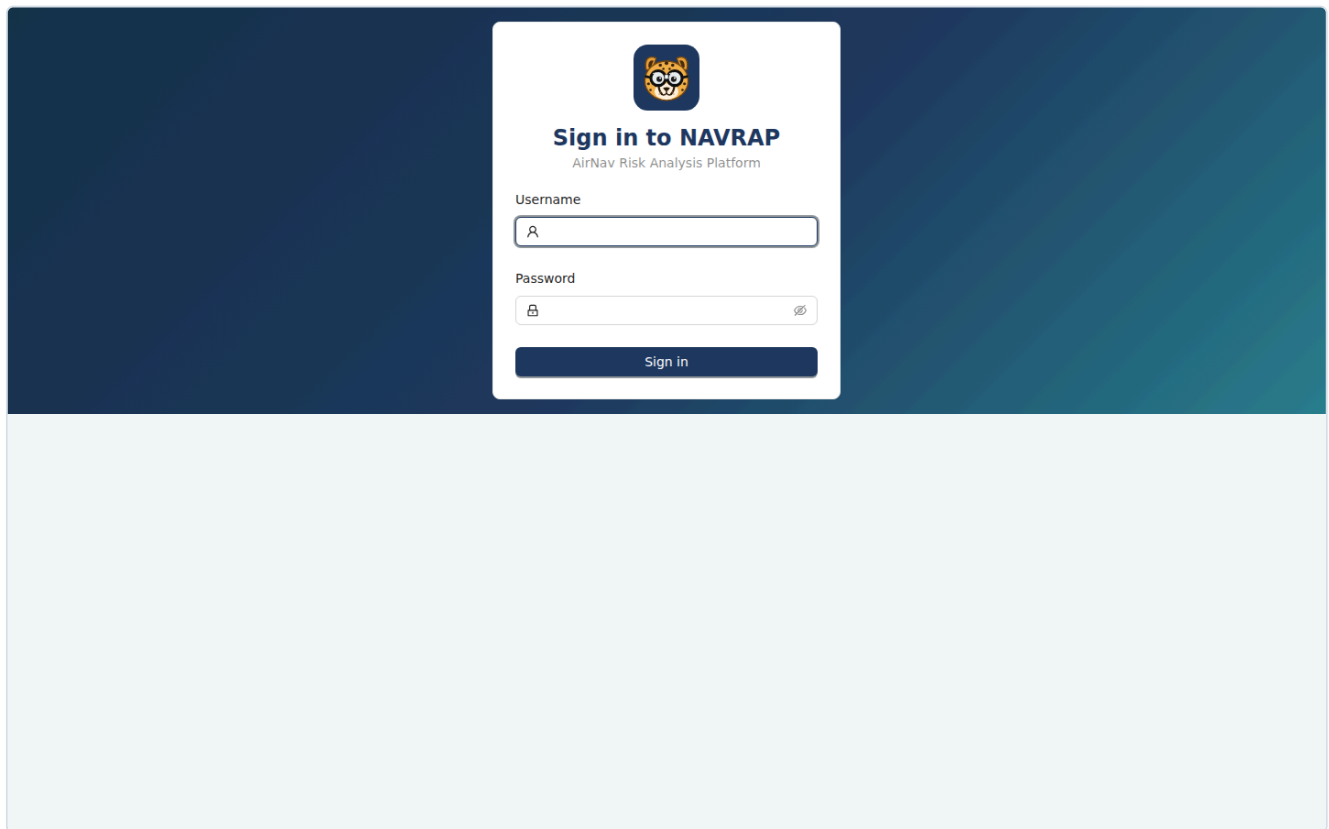
NAVRAP is the web application that implements the *AirNav Risk Analysis Manual*. It lets AirNav staff plan, run, review and approve safety risk assessments with twenty-six methods, keeps every hazard in one hazard log, and produces the Safety Assessment Report.

This guide explains how to use the application. For *why* and *when* to use each method, see the Manual ([docs/AirNav_Risk_Analysis_Manual.pdf](#)). For the formal requirements, see the SRS ([docs/AirNav_Risk_Analysis_Software_Requirements.pdf](#)).

1. Getting started

1.1 Signing in

Open NAVRAP in a browser (Chrome, Edge, Firefox or Safari) at the address your administrator gives you — for a local installation this is `http://localhost:8080` . Enter your username and password.



Sign-in page

1.2 Roles

What you can do depends on your role:

Role	Can do
Viewer	Read everything in their scope; update actions they own.
Assessor	Create projects, assessments and method studies; edit worksheets and diagrams; send hazards to the hazard log; raise actions; submit assessments for review.
Reviewer	Everything an assessor can, plus return, endorse or reject assessments, and read the audit trail.
Authority	Accept (or reject) residual risk, but only for the risk regions assigned to them (for example <i>Tolerable (lower)</i>).
Admin	Manage users, roles and the risk classification scheme. Admin is not a substitute for the proper acceptance authority in normal use.

1.3 Demo accounts

When NAVRAP is installed with demo data (`ARAP_SEED_DEMO=true`), these accounts exist, all with password `demo1234` :

Username	Role	May accept
assessor	Assessor	—
reviewer	Reviewer	—
director	Authority	Tolerable (lower), Acceptable
accexec	Authority (Accountable Executive)	Tolerable (upper), Tolerable (lower), Acceptable
viewer	Viewer	—

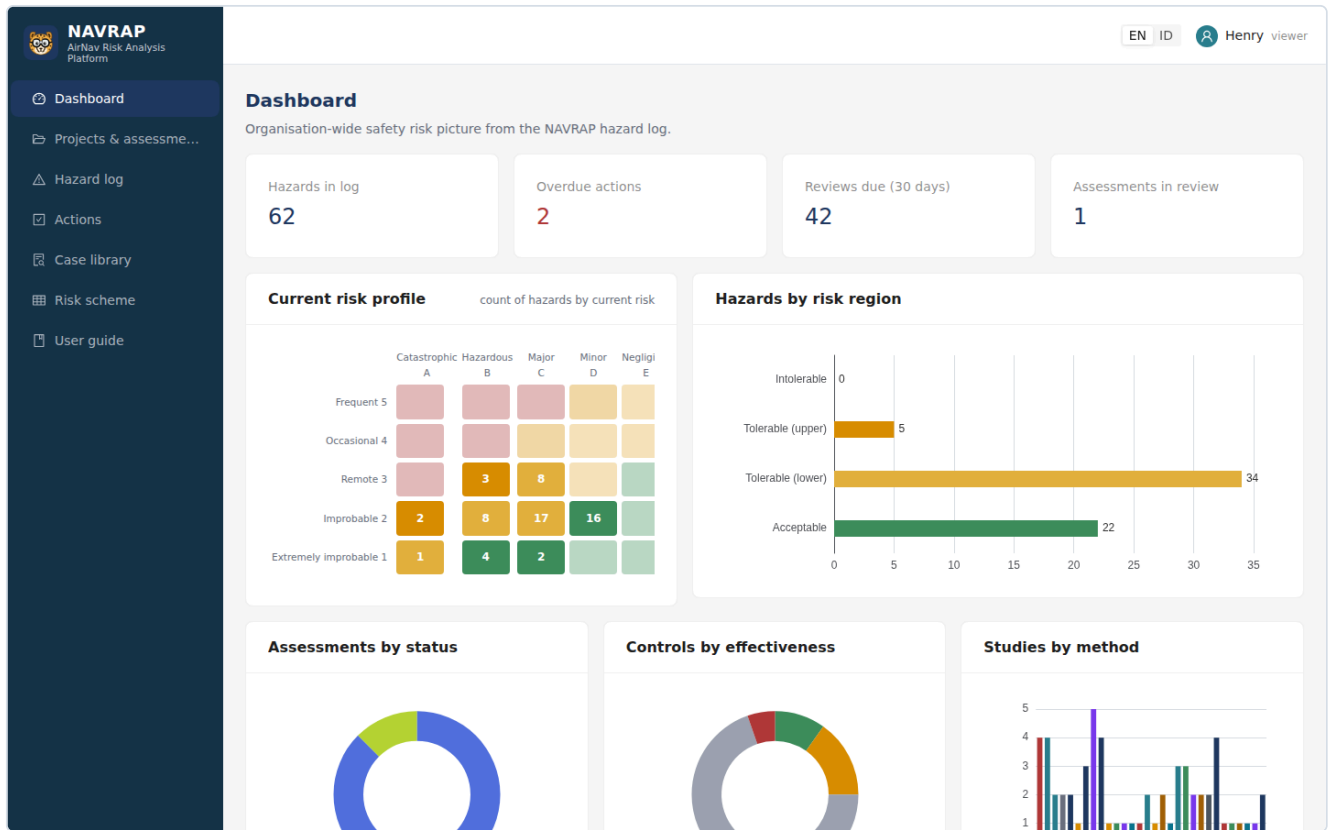
The administrator account is `admin` ; its password is set at installation. **Disable the demo data and change all passwords before real use.**

1.4 Screen layout and language

- The **left menu** gives access to the Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme and User guide (plus Users and Audit trail for administrators and reviewers).
- The **EN / ID** switch at the top right changes the interface language between English and Bahasa Indonesia. Your choice is remembered on that computer. Text you type into worksheets is stored as entered.
- Your name and role are shown top right; click them to sign out.

2. The dashboard

The dashboard is the organisation-wide risk picture, built from the hazard log.



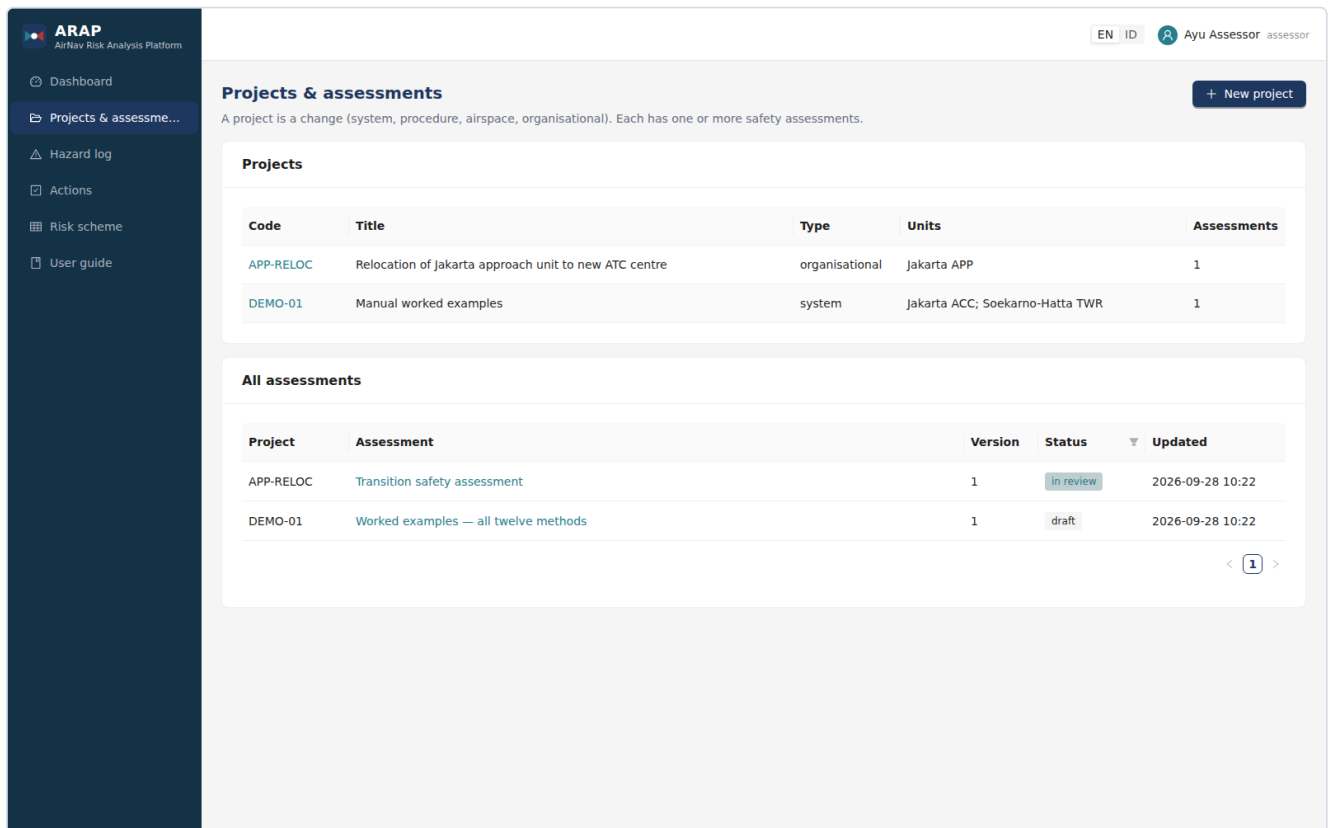
Dashboard

- **Key figures** — hazards in the log, overdue actions, hazard reviews due within 30 days, and assessments waiting for review.
- **Current risk profile** — the 5 × 5 risk matrix with the number of hazards in each cell (residual risk where rated, otherwise initial risk).
- **Hazards by risk region, Assessments by status, Controls by effectiveness** and **Studies by method** charts.
- **Overdue actions** and **Reviews due** lists — click an item to open it.

3. Projects and assessments

3.1 Projects

A **project** represents one change: a new system, a procedure, an airspace change, an organisational change or a temporary change. Go to **Projects & assessments** → **New project** and enter a code (e.g. ADSB-2027), a title, the type of change, the units affected and the sponsor.



ARAP
AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

Projects & assessments [+ New project](#)

A project is a change (system, procedure, airspace, organisational). Each has one or more safety assessments.

Projects

Code	Title	Type	Units	Assessments
APP-RELOC	Relocation of Jakarta approach unit to new ATC centre	organisational	Jakarta APP	1
DEMO-01	Manual worked examples	system	Jakarta ACC; Soekarno-Hatta TWR	1

All assessments

Project	Assessment	Version	Status	Updated
APP-RELOC	Transition safety assessment	1	in review	2026-09-28 10:22
DEMO-01	Worked examples — all twelve methods	1	draft	2026-09-28 10:22

< 1 >

Projects

3.2 Assessments

Open the project and choose **New assessment**. Describe the **scope and system boundary**, the **operational environment** and your **assumptions** (Manual §2.3). The assessment records which version of the risk scheme it uses.

The assessment page brings everything together:

Worked examples — all twelve methods draft v1

Submit for review | Report (.docx)

Scope, environment and assumptions Edit

Scope: Demonstration of every ARAP method module using the Manual's examples.
 Environment: Illustrative values only.
 Assumptions: All numbers are illustrative (Manual, How to use).
 Risk scheme: version 1

Method studies (12) Add study

Suggested for a system change (Manual §4.1): HAZID FHA FMEA / FMECA FTA Bowtie analysis

Method	Study	Status	Updated
Bowtie analysis	Runway incursion bowtie (Manual §6)	open	2026-09-28 10:22
HAZID	HAZID — approach unit relocation (Manual §7.4)	open	2026-09-28 10:22
HAZOP	HAZOP — estimate message coordination (Manual §8.3)	open	2026-09-28 10:22
JHA	JHA — obstruction light on radar tower (Manual §9.3)	open	2026-09-28 10:22
FMEA / FMECA	FMEA — sector VHF air-ground (Manual §10.3)	open	2026-09-28 10:22
LOPA	LOPA — level bust (Manual §11.3)	open	2026-09-28 10:22
FHA	FHA — ADS-B en-route surveillance (Manual §12.3)	open	2026-09-28 10:22
STPA	STPA — descent clearances (Manual §13.2)	open	2026-09-28 10:22
FTA	FTA — loss of surveillance picture (Manual §14.3)	open	2026-09-28 10:22
Fatigue modeling	Fatigue — night shift with/without nap (Manual §15.4)	open	2026-09-28 10:22
FRAM	FRAM — descent clearances in a busy arrival sector (Manual §16)	open	2026-09-28 10:22
Bayesian belief network	BBN — coordination error and loss of separation (Manual §17.3)	open	2026-09-28 10:22

Hazards (5) Open hazard log →

Ref	Hazard	Initial	Residual	Controls
HZ-0001	Unauthorised entry onto active runway	3A	2A	2
HZ-0002	Level bust leading to potential mid-air collision	3A	3A	2
HZ-0003	Total loss of surveillance picture at ACC sector	3B	3B	1
HZ-0004	ATCO fatigue at end of night shift	4C	3C	1
HZ-0005	New site outside radio line of sight of low-level sector	3B	—	0

Risk acceptance

Worst current/residual region: Tolerable (upper)
 Acceptance authority: Accountable Executive

Actions (5) All actions →

- ACT-0001 Install diverse, physically separate surveillance data path
CNS engineering - due 2026-12-27 - open
- ACT-0002 Implement SFL mismatch alert in ATM system
ATIS systems - due 2027-02-25 - open
- ACT-0003 Collect KSS survey for two roster cycles
FRMS specialist - due 2026-09-18 - open
- ACT-0004 Audit stop-bar U/S procedure compliance
Aerodrome ops - due 2026-09-25 - in progress
- ACT-0005 Radio coverage study for new site
CNS engineering - due 2026-10-28 - open

Review and approval history

Not yet submitted.

Assessment page

- **Method studies** — the analyses carried out for this assessment. NAVRAP suggests methods for the type of change (Manual §4.1); the suggestion is only advice.
- **Hazards** — hazard-log entries that belong to this assessment, with initial and residual risk.
- **Risk acceptance** — the worst residual risk region and the authority that must accept it.
- **Actions** and the **review and approval history**.
- Workflow buttons at the top right (see section 15) and **Report (.docx)**.

3.3 Adding a method study

On the assessment page choose **Add study**, pick the method (suggested methods are marked ★) and give the study a title, e.g. *HAZOP of cut-over plan*. The study opens straight away.

3.4 Starting from an assessment template

The screenshot displays the ARAP (AirNav Risk Analysis Platform) interface. A modal window titled "New safety assessment" is open, allowing the user to create a new assessment from a template. The "Start from (optional)" dropdown is set to "New AIM system acquisition (installation → transition)". An information box indicates that 14 pre-filled studies will be created, listing various analysis methods: FHA, HAZOP, FMEA, FTA, CCA, STPA, HTA, HRA, SEC, JHA, SWIFT, BOWTIE, SIM, and GSN. The modal includes text boxes for the following fields:

- Title:** Safety assessment — new AIM system acquisition
- Scope and system boundary (Manual §2.3):** Acquisition, installation, data migration and transition into operation of a new Aeronautical Information Management (AIM) system replacing the legacy AIS system. Functions in scope: receipt and registration of raw data from originators; validation and verification (incl. CRC-...)
- Operational environment:** Aeronautical information for all Indonesian FIRs, published to the AIRAC 28-day cycle; NOTAM office operating H24; data integrity classification per ICAO Annex 15 and PANS-AIM
- Assumptions:** A-1 The vendor provides software assurance evidence (EUROCAE ED-153 or equivalent) for the assigned assurance level. A-2 The legacy AIS system remains available as a hot-

A "Create" button is located at the bottom left of the modal. The background shows the ARAP dashboard with a sidebar menu and a main content area with a "New assessment" button and a table of studies.

New assessment from a template

In **New assessment**, the field **Start from** offers ready-made assessment templates. When you choose one:

- NAVRAP fills in the title, scope, environment and assumptions. Edit them for your change.
- On **Create**, NAVRAP also creates all of the template's method studies, pre-filled with proposed content.
- Nothing is rated for you. Likelihoods, calculations and hazard-log entries are left for your workshops, and severities are proposals to confirm.

New AIM system acquisition (Manual Appendix C) creates 14 studies covering the whole life of the change, from specification through installation and data migration to transition:

Study	Pre-filled content
FHA	8 AIM functions and 12 failure conditions (loss, delay, detected and undetected error) with proposed severities and requirements
Data-flow HAZOP	7 nodes from originator to user, including legacy → new migration; 15 deviations filled in (use Deviations to add the rest)
FMEA	8 architecture items: servers, fail-over, database, CRC module, validation rules, AMHS gateway, time source, eAIP web
FTA	"Erroneous critical data published undetected", with an entry-error branch and a storage/transfer corruption branch; data entry and verification come from the HRA
CCA, STPA, HTA, HRA, security	Independence claims, the officer-automation control loop, the NOTAM publication task, data-entry and verification error probabilities, and integrity/access threats
JHA, SWIFT	Installation works beside the live legacy AIS, and what-ifs for installation and cut-over
Bowtie	"Loss of, or erroneous, AIS during transition", with migration, testing, training, interface, timing and installation threats, plus rollback and manual-NOTAM recovery
Simulation	The shadow-operation plan and acceptance criteria (zero critical-data differences; processing time; rejections)
GSN	The argument G0-G7, with every solution already linked to the new studies

Demo project **DEMO-04** is this template filled in with illustrative values. It has likelihoods, a calculated fault tree (2.86×10^{-5} per critical item, which is above the objective, so a further independent check is needed), shadow-operation results with one criterion not met, 7 hazards and 6 actions.

Major organisational change (Manual Appendix D) creates 11 studies covering planning, execution and post-implementation:

ARAP
AirNav Risk Analysis Platform

Dashboard
Projects & assessments...
Hazard log
Actions
Risk scheme
User guide

EN ID Ayu Assessor assessor

Organisational function map
Organisational function map — current to new owner

Safety assessment — major organisational change · Manual chapter App. D · template v1.0
4 participants
Saved

+ Add row
Delete selected
Send all rows to hazard log
Double-click a cell to edit. Tick rows to select.

ID		Safety-related function	Gaps	Category	Current owner	New owner	Comp...	Capacity	Authority
FN-01	☐	Accept residual risk in the Tolerable (lower) region for branch changes	authority pending	SMS	Director of Operations	Regional Operations Director	confirmed	adequate	pending
FN-02	☐	Own and review branch hazard-log entries	capacity stretched	SMS	Branch Safety Officer	Regional S&Q Unit	confirmed	stretched	in place
FN-03	☐	Triage occurrence reports (ERC) within 72 h	capacity overloaded	SMS	Branch Safety Officer	Regional S&Q Unit	confirmed	overloaded	in place
FN-04	☐	Investigate branch occurrences	competence capacity stretched	SMS	Branch Safety Officer (trained investigator)	Regional S&Q Unit	gap	stretched	in place
FN-05	☐	Safety Manager reports directly to the Accountable Executive	authority none no handover	SMS	Safety Manager -- Accountable Executive	Regional S&Q Unit -- Regional Operations Director	confirmed	adequate	none
FN-06	☐	Facilitate safety assessments of branch changes	capacity stretched	SMS	Branch Safety Officer	Regional S&Q Unit	confirmed	stretched	in place
FN-07	☐	Supervise the APP watch	capacity stretched	Operations	APP Watch Supervisor	Operations Supervisor (TWR + APP)	confirmed	stretched	in place
FN-08	☐	Supervise the TWR watch	capacity stretched	Operations	TWR Watch	Operations	confirmed	stretched	in place

18 functions mapped · 13 with gaps · 2 without a new owner · 2 still to assess
Every row needs a named new owner with confirmed competence, adequate capacity, authority in place and a planned handover before go-live. Send rows with gaps to the hazard log.

Organisational function map

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Study	Pre-filled content
Organisational function map	18 safety-related functions: SMS roles, supervision, training and competence, technical functions and bowtie barrier owners, each with current owner → new owner. Competence, capacity, authority and handover start as <i>unknown</i> . The Gaps column flags orphans (no new owner), competence gaps, stretched or overloaded capacity, pending authority, missing handover and anything still to assess. Rows with gaps can be sent to the hazard log.
FHA	Failure conditions of the organisational functions: triage delayed, investigations degraded, safety function not independent, merged supervisor overloaded, LoAs without an owner, and so on
STPA × 2	The current and the new control structure, so you can compare the feedback loops. The new one includes unsafe control actions (UCAs) and loss scenarios.
HAZID	Uses its own organisational guidewords : roles, authority, competence, workload, interfaces, communication, corporate memory, contractors, culture, transition. Coverage is counted against this list.
HTA	The merged Operations Supervisor post, with a plan that sets priorities between watch, OJT and events
Delphi	The question and panel for the likelihood of triage failure after transfer
SWIFT, bowtie	What-ifs for the transition plan; the top event "safety-related function not performed, or performed without competence, capacity or authority", with phased go/no-go and reversion
SPI register	10 indicators with target, alert level, owner and escalation. Rows at red are listed at the top.
GSN	G0–G7, linked to all the studies above

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Dashboard
Projects & assessments...
Hazard log
Actions
Risk scheme
User guide

SPI register
SPI register — post-implementation monitoring

Safety assessment — major organisational change · Manual chapter 32 · template v1.0
4 participants
Saved

+ Add row
Delete selected
Double-click a cell to edit. Tick rows to select.

ID	Indicator	Type	Barrier / hazard	Base...	Target ...	Alert level	Frequ...	Owner	Current	Status
S-01	☐ Vacant or acting posts among safety-critical functions	leading	Function map	0	0	≥ 2	monthly	HR	1	amber
S-02	☐ Function-map rows with gaps (owner, competence, capacity, authority, handover)	leading	Function map	18 to assess	0 at go-live	≥ 1 at go-live	weekly	Transition board	13	red
S-03	☐ Overdue hazard-log actions and reviews (%)	leading	SMS hazard log	6%	≤ 5%	> 15%	monthly	Regional S&Q	9%	amber
S-04	☐ Occurrence reports triaged within 72 h (%)	leading	FHA triage	97%	≥ 95%	< 85%	monthly	Regional S&Q	88%	amber
S-05	☐ Voluntary safety reports per month vs baseline	leading	Reporting culture	24 / month	≥ baseline	drop > 30%	monthly	Safety Manager	19	amber
S-06	☐ Trainees waiting for OJT allocation	leading	OJT coordination	2	≤ 3	rising 2 months	monthly	Terminal Unit Manager	3	green
S-07	☐ Overtime hours per ATCO per month	leading	Fatigue / roster	8 h	≤ 10 h	> 12 h	monthly	Terminal Unit Manager	11 h	amber
S-08	☐ Median time to close an investigation (days)	lagging	Investigation	28	≤ 30	> 60	quarterly	Regional S&Q	41	amber
S-09	☐ Safety culture survey index (0–100)	leading	Culture	68	≥ 68	drop ≥ 5 points	6-monthly	Safety Manager	—	
S-10	☐ Occurrences with non-communicational	leading	Investigations	2 / quarter	≤ baseline	rising 2	quarterly	Regional S&Q	2	amber

At alert level: S-02 — apply the agreed escalation

SPI register

Demo project **DEMO-05** is this template filled in for an illustrative branch reorganisation. It has 13 of 18 functions with gaps (2 orphans), the Safety & Quality function losing its direct line to the Accountable Executive, a Delphi estimate converging on 0.12, SPI values, 7 hazards and 6 actions.

The same modules can be added to any assessment on their own: **Organisational function map** and **SPI register** appear in **Add study**.

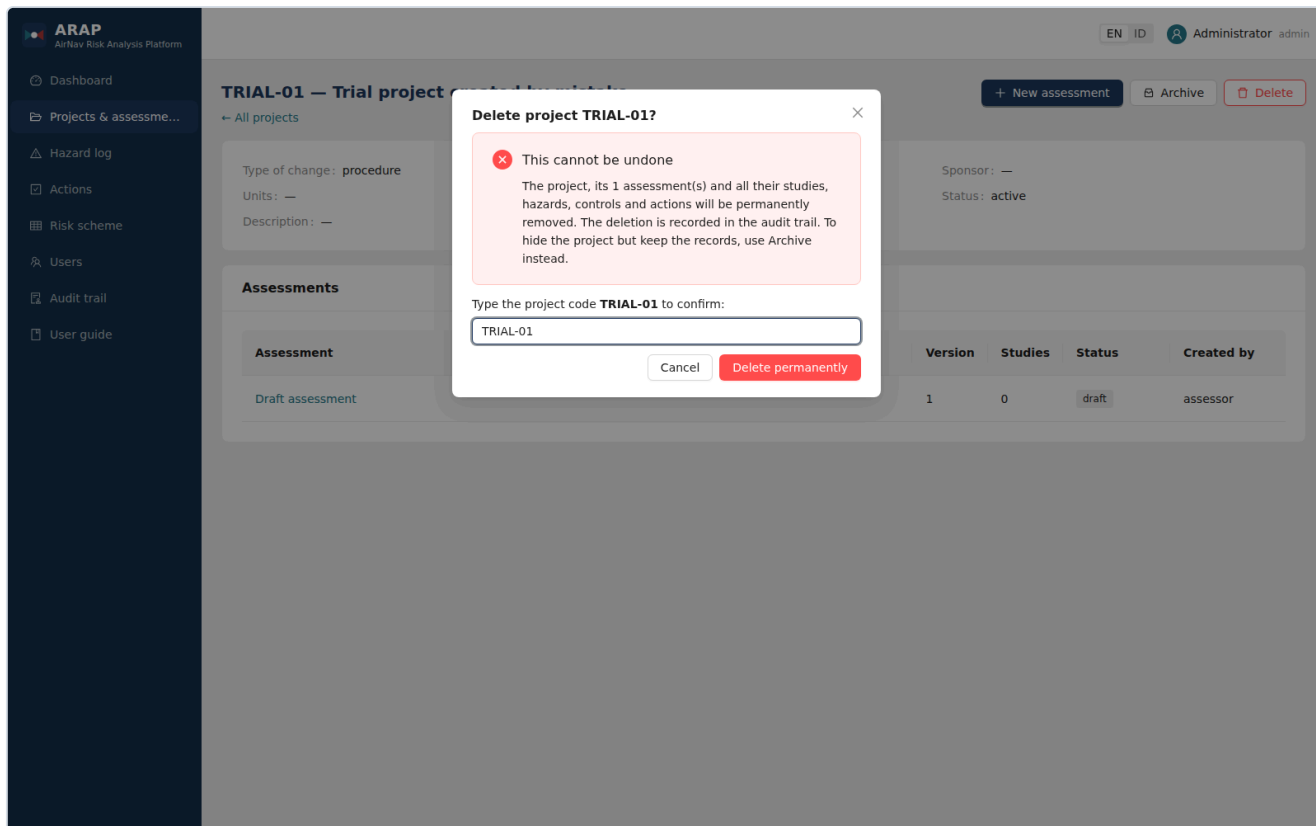
3.5 Archiving and deleting a project

A project you no longer need can be **archived** or, by an administrator, **deleted**. The buttons are at the top right of the project page.

	Archive	Delete
Who	Assessors, reviewers and administrators	Administrators only
What happens	The project and its assessments disappear from the project list and the assessment list. Its hazards and actions stay in the hazard log, the action list and the dashboard, because they may still be live. Every record is kept and can still be opened by its link. No new assessments can be added.	The project and all its assessments, studies, hazards, controls and actions are removed permanently.
Undo	Restore on the project page	Not possible
When refused	—	If any assessment has been endorsed, accepted, closed or superseded. Those are safety records; archive the project instead.
Audit trail	"archive" / "restore"	"delete", with the project details and the number of records removed

To archive: open the project and click **Archive**. A yellow banner shows that the project is archived. To find it again, switch on **Show archived** above the project list; archived projects carry an *archived* tag. Open it and click **Restore** to make it active.

To delete: sign in as an administrator, open the project and click **Delete**. Type the project code exactly (for example TRIAL-01) and click **Delete permanently**. The button stays disabled until the code matches. NAVRAP then returns to the project list and reports how many assessments, studies, hazards and actions were removed.



Deleting a project

Use delete for projects created by mistake, test projects and the demo projects. A deleted demo project is not re-created when NAVRAP restarts. For a real change that has been cancelled or completed, archive it: the assessment is part of the safety record.

4. Working in a method study

Every study page has the same header:

- The **method** tag and the **study title** (editable).
- A link back to the assessment, the Manual chapter for the method and the template version.
- The number of **participants** recorded.
- **Save**. Changes are not saved automatically: the button reads *Save* while there are unsaved changes and *Saved* afterwards. If you try to leave the page with unsaved changes NAVRAP asks you to confirm.

When the assessment is endorsed, accepted or closed, or if your role cannot edit, the study opens **read-only** and a blue banner says so.

4.1 Sending results to the hazard log

Every method has a **Send to hazard log** button. It creates a hazard-log entry for each selected worksheet row (or for the bowtie as a whole, or for each STPA hazard), carrying over the causes, consequences, controls and any severity and likelihood. Sending the same row again **updates** the existing hazard rather than creating a duplicate. A message lists the hazard references created, e.g. HZ-0007, HZ-0008 . Rate the risk and complete the controls in the [hazard log](#).

5. Worksheet methods

HAZID, HAZOP, JHA, FMEA/FMECA and FHA use a spreadsheet-style worksheet.

Editing the worksheet

- **Double-click** a cell (or press Enter) to edit it; press Enter or click elsewhere to finish.
- Columns with fixed values — severity, likelihood, guideword, control type, failure type — show a drop-down list.
- **Add row** appends a row with the next ID (HZ-06 , HP-04 , ...).
- Tick the checkboxes on the left to select rows, then **Delete selected** or **Send ... to hazard log**.
- Click a column header to sort; use the filter icon to filter.
- The **Risk** column is computed from severity and likelihood using the current risk matrix.

The screenshot displays the ARAP AirNav Risk Analysis Platform interface. On the left is a dark sidebar with navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme, and User guide. The main area shows a HAZID worksheet titled 'HAZID — approach unit relocation (Manual §7.4)'. At the top right of the main area, it says 'EN ID' and 'Ayu Assessor assessor'. Below the title, there's a note 'Worked examples — all twelve methods · Manual chapter 7 · template v1.0' and '2 participants' and 'Saved'. The worksheet table has columns: ID, Guideword, Hazard, Cause(s), Consequence, Existing controls, S, L, Risk, and Actions. There are five rows of data. The first and fifth rows are selected, indicated by checkboxes in the first column. Below the table is a 'Guideword coverage' panel with a bar chart showing the count of rows for each guideword: People: 1, Procedures: 0, Equipment / systems: 1, Airspace and aerodrome: 0, Traffic: 0, Environment: 1, Interfaces: 1, Transition: 1, Organisation: 0. There is also a dropdown menu to 'Record "no hazard identified" for a guideword'.

ID	Guideword	Hazard	Cause(s)	Consequence	Existing controls	S	L	Risk	Actions
HZ-01	Transition	Loss of surveillance display at cut-over	Data feed not re-routed; configuration error	Loss of separation during cut-over	Cut-over at low traffic; old site on hot standby	C	3	3C	HAZOP of cut rollback criteria
HZ-02	People	ATCOs unfamiliar with new CWP HMI	Short training window	Delayed or erroneous inputs	Simulator training	D	4	4D	Minimum hours before live ops
HZ-03	Interfaces	Wrong telephone coordination lines to TWR	VCCS mapping error	Late coordination of missed approach	Test plan	C	3	3C	End-to-end line each TWR
HZ-04	Equipment / systems	Single power feed to new building	Design	Total loss of service	UPS 30 min	B	2	2B	Second feed of power
HZ-05	Environment	New site outside radio line of sight of low-level sector	Topography	Loss of VHF contact below FL050	None yet	B	3	3B	Coverage study site

Guideword coverage

People: 1 Procedures: 0 Equipment / systems: 1 Airspace and aerodrome: 0 Traffic: 0 Environment: 1 Interfaces: 1 Transition: 1 Organisation: 0

Record "no hazard identified" for a guideword

HAZID worksheet after sending two rows to the hazard log

5.1 HAZID (Manual §7)

Work through the guidewords (People, Procedures, Equipment/systems, Airspace and aerodrome, Traffic, Environment, Interfaces, Transition, Organisation). The **Guideword coverage** panel below the worksheet shows how many rows each guideword has. When a guideword produces nothing, record that explicitly with **Record "no hazard identified"**. Use the *Needs method* column to flag rows that need a detailed method.

5.2 HAZOP (Manual §8)

1. In **Nodes and design intent**, add a node per procedure step or message exchange (+ **Node**), choose its type (*procedural* or *data-flow*) and write the design intent.
2. Press **Deviations** on a node. Choose the parameters and guidewords; NAVRAP adds one worksheet row per combination (e.g. *Message — No / Not*).

- For each row either tick **N/A** (not meaningful) or record causes, consequences, safeguards, S, L and a recommendation.
- The **Progress** column counts rows analysed out of rows generated.

Nodes and design intent

ID	Node	Type	Design intent	Progress
N3	Node 3: EST/CPL message upstream → downstream	data-flow	Downstream sector receives complete, correct and timely data for every flight crossing the boundary.	3/3 analysed Deviations

+ Add row Delete selected Send all rows to hazard log Double-click a cell to edit. Tick rows to select.

ID	Node	Parameter	Guideword	Deviation	Possible causes	Consequences	Safeguards	S	L
HP-01	N3	Message	No / Not	Estimate not received	Link failure; flight plan not activated; address error	Unknown traffic enters sector	Link-status alarm; verbal coordination backup	C	3
HP-02	N3	Content	Other than	Coordinated level differs from actual	Level changed after EST; no revision	Level conflict downstream	Mode C/S check; STCA	B	3
HP-03	N3	Timing	Late	Estimate inside agreed time	Short-notice route change	Workload	LoA fallback; verbal	D	4

HAZOP worksheet with node panel

5.3 JHA (Manual §9)

Fill in the job header (job, location, permits), then one row per job step with its hazards, controls, control type and the person responsible. NAVRAP warns when a step relies only on **Administrative** controls or **PPE**, the lowest levels of the hierarchy of controls.

5.4 FMEA / FMECA (Manual §10)

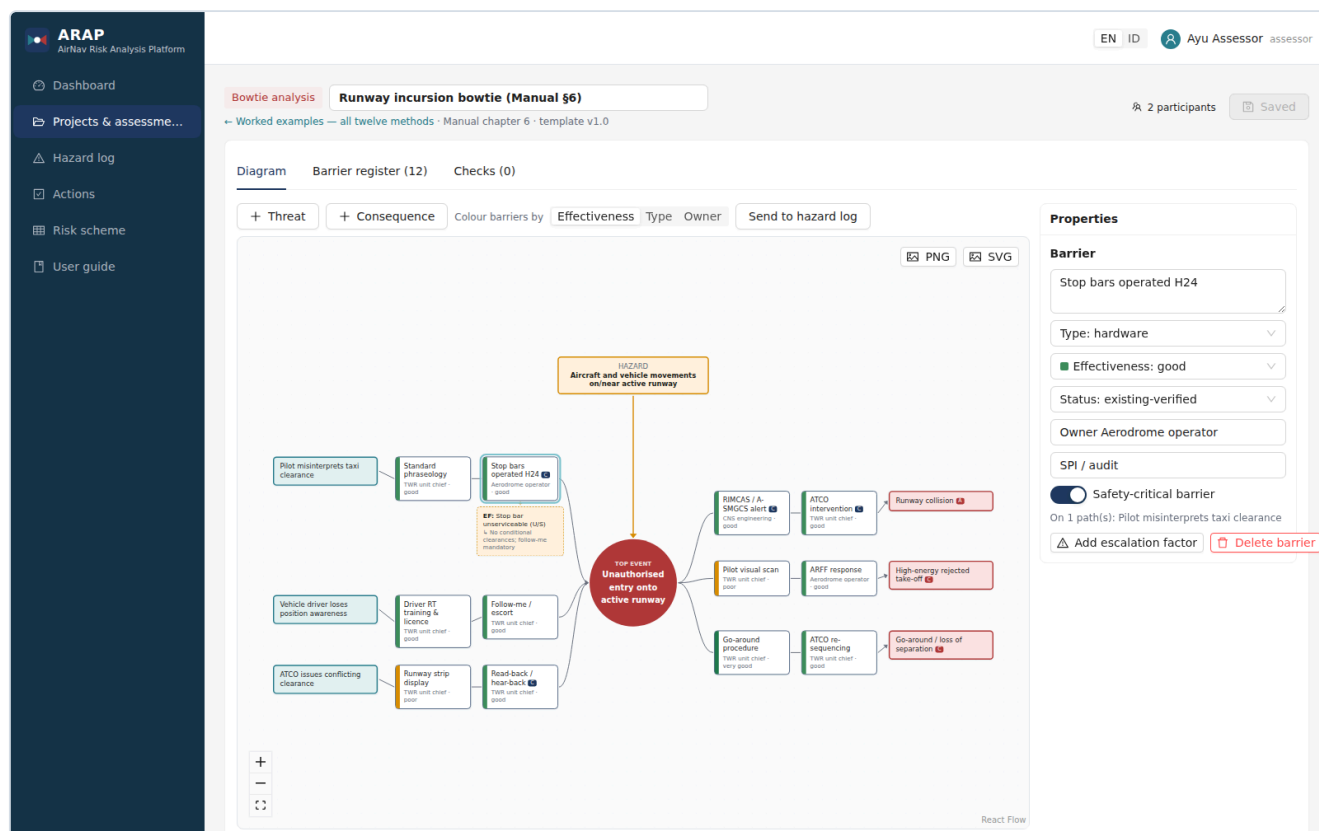
One row per failure mode: item, failure mode, cause, end effect on the ATS, detection, compensation, and **S, O, D** ratings (1–10). NAVRAP computes the **RPN** ($S \times O \times D$), colours high values and sorts by RPN. It also lists high-severity modes ($S \geq 8$) so they are not missed because of a low RPN. For FMECA, add the failure rate λ_p , mode ratio α , conditional probability β and operating time t .

5.5 FHA (Manual §12)

One row per failure condition: the function, the failure type (total loss, partial loss, erroneous detected/undetected, unintended, delayed), the condition, the operational effect and the **severity**. The **Safety objective** column is computed from the risk scheme: the maximum frequency per operating hour that keeps the failure condition at or better than *Tolerable (lower)* (Manual Table 12.1). Record the derived safety requirements in the last column.

6. Bowtie analysis

The bowtie editor draws the diagram automatically: threats on the left, the top event in the centre, consequences on the right, barriers in order on each line and escalation factors below the barrier they weaken.



Bowtie editor with a barrier selected

Building the bowtie

1. Click the **HAZARD** box and the **TOP EVENT** circle to edit their text. Phrase the top event as the moment control is lost, *before* any damage.
2. **+ Threat** and **+ Consequence** add lines. Click one to edit its text (and a consequence's severity).
3. With a threat or consequence selected, the **Properties** panel lists its barriers in order: - **New barrier on this path** adds a barrier; - **Add an existing (shared) barrier** puts a barrier that already exists on this line too — a shared barrier shows **xn**; - the arrows reorder barriers and the bin removes a barrier from this line only.
4. Click a **barrier** to set its type, effectiveness, status (existing-verified, existing-unverified, planned), owner, SPI/audit reference and whether it is **safety-critical** (C badge).
5. With a barrier selected, **Add escalation factor** attaches a condition that degrades it. Select the escalation factor to write its escalation-factor barriers, one per line.

Viewing

- **Colour barriers by** effectiveness (default), type or owner.
- Zoom with the mouse wheel or the **+ / –** controls; drag the background to pan; the frame icon fits the diagram.
- **PNG** and **SVG** download the full diagram as an image.
- **Barrier register** lists every barrier with its attributes (Manual §6.3).

- **Checks** lists issues: a top event phrased like a consequence, lines with fewer than two barriers, barriers without an owner.

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Bowtie analysis

Runway incursion bowtie (Manual §6)

Worked examples

all twelve methods

Manual chapter 6

template v1.0

2 participants

Saved

Diagram

Barrier register (12)

Checks (0)

Barrier	Side	Type	Owner	Effectiveness	Critical	Status	Paths	SPI / audit
Standard phraseology	prevention	human	TWR unit chief	good		existing-verified	Pilot misinterprets taxi clearance	
Stop bars operated H24	prevention	hardware	Aerodrome operator	good	Yes	existing-verified	Pilot misinterprets taxi clearance	
Driver RT training & licence	prevention	human	TWR unit chief	good		existing-verified	Vehicle driver loses position awareness	
Follow-me / escort	prevention	human	TWR unit chief	good		existing-verified	Vehicle driver loses position awareness	
Runway strip display	prevention	human-hardware	TWR unit chief	poor		existing-verified	ATCO issues conflicting clearance	
Read-back / hear-back	prevention	human	TWR unit chief	good	Yes	existing-verified	ATCO issues conflicting clearance	
RIMCAS / A-SMGCS alert	recovery	hardware	CNS engineering	good	Yes	existing-verified	Runway collision	
ATCO intervention	recovery	human	TWR unit chief	good	Yes	existing-verified	Runway collision	
Pilot visual scan	recovery	human	TWR unit chief	poor		existing-verified	High-energy rejected take-off	
ARFF response	recovery	human-hardware	Aerodrome operator	good		existing-verified	High-energy rejected take-off	
Go-around procedure	recovery	human	TWR unit chief	very good		existing-verified	Go-around / loss of separation	
ATCO re-sequencing	recovery	human	TWR unit chief	good		existing-verified	Go-around / loss of separation	

Barrier register

7. LOPA

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Scenario Level bust leading to a potential mid-air collision

Initiating event Level bust in sector X

f(IE) 0.5

Unit per year

Data source Illustrative

Tolerable target 0.00001

Enabling conditions / conditional modifiers

+ Add

Condition	Probability
Conflicting traffic present	0.05

Safeguards — credited only if all four IPL criteria are met

+ Add

Safeguard	PFD	indep.	effec.	depen.	audit.	Shared dependencies	Justi
ATCO monitoring and intervention	0.10	☒	☒	☒	☒	surveillance	
STCA alert and ATCO resolution	0.10	☒	☒	☒	☒	surveillance	
ACAS II RA followed	0.10	☒	☒	☒	☒	e.g. surveillance	
SFL mismatch alert (proposed)	0.10	☒	☒	☐	☒	e.g. surveillance	
Recurrent RT training	0.10	☐	☐	☒	☒	e.g. surveillance	

Calculate

Mitigated frequency

2.50e-5 per year

Target

1.00e-5 per year

Verdict

Gap x2.5

Further reduction needed

x2.5 (0.40 orders of magnitude)

Credited IPLs

ATCO monitoring and intervention; STCA alert and ATCO resolution; ACAS II RA followed

Not credited: SFL mismatch alert (proposed)

Missing: dependable

Not credited: Recurrent RT training

Missing: independent, effective

IPLs ATCO monitoring and intervention, STCA alert and ATCO resolution share dependency 'surveillance' — independence must be justified

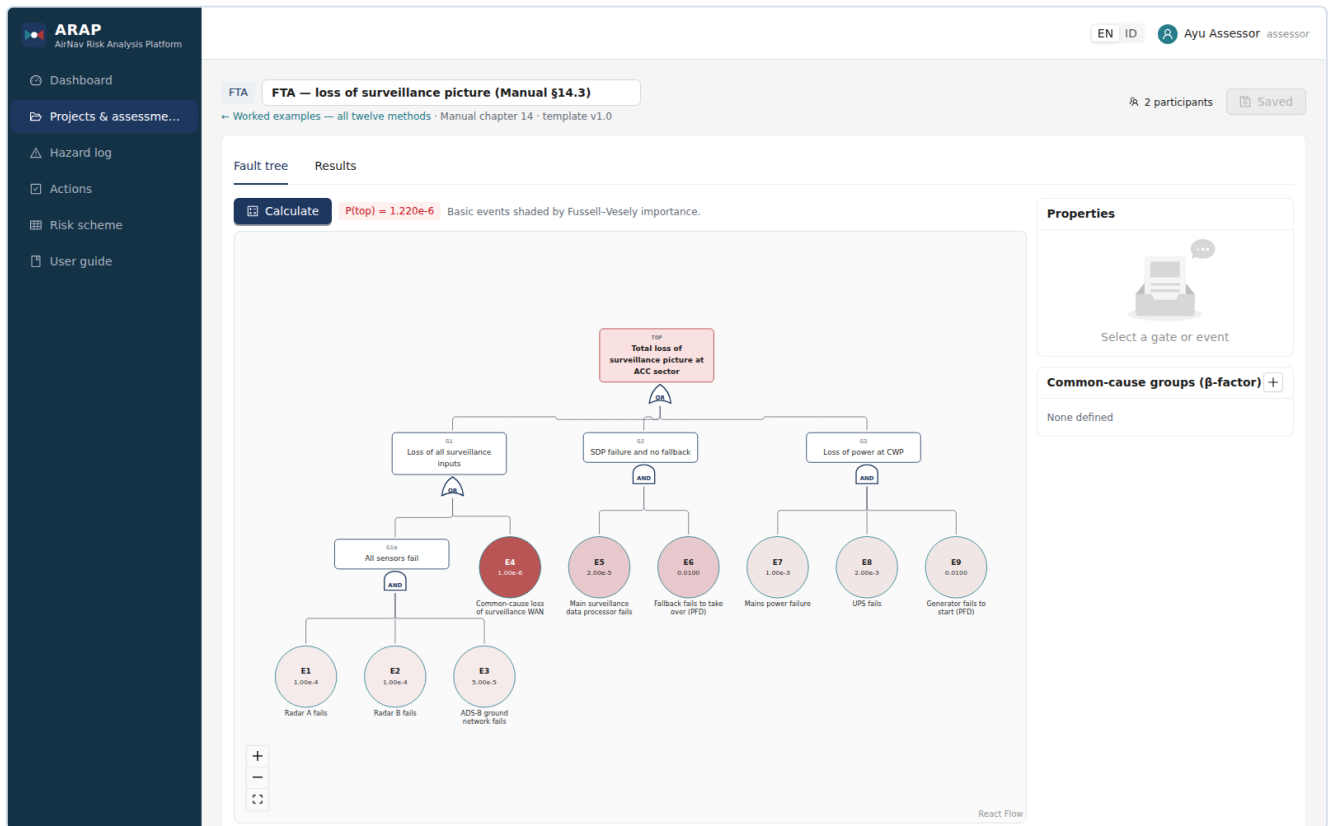
Layer	Frequency (per year)
Initiating event	5.0e-1
Conflicting traffic present	2.5e-2
ATCO monitoring and intervention	2.5e-3
STCA alert and ATCO resolution	2.5e-4
ACAS II RA followed	2.5e-5

LOPA calculator

- Describe the **scenario**, the **initiating event**, its frequency, unit and data source, and the **tolerable target** frequency.
- Add **enabling conditions / conditional modifiers** with their probabilities.
- Add **safeguards** with a probability of failure on demand (PFD). Tick each of the four IPL criteria — *independent, effective, dependable, auditable* — only if it is justified, and write the justification. A safeguard is credited **only if all four are ticked**.
- Record shared dependencies (e.g. `surveillance`) — NAVRAP warns if two credited layers share one.
- Press **Calculate**. The result shows the mitigated frequency, the verdict (*Target met* or *Gap x n*), the extra risk reduction needed, the safeguards not credited and why, and a log-scale chart of the frequency after each layer.

In the demo study the proposed *SFL mismatch alert* is not yet ticked *dependable*, so the result shows a gap of x2.5. Tick it and recalculate to see the target met (2.5×10^{-6} per year).

8. Fault tree analysis



Fault tree

Building the tree

- Click a gate and use **OR gate**, **AND gate**, **Basic event** or **Undeveloped** in the Properties panel to add inputs. **Reuse an existing node as input** links a node already in the tree (e.g. a shared support system).
- For a **vote** gate set k (k -out-of- n).
- For a **basic event** give either a probability p , or a failure rate λ with an exposure time t ($P = 1 - e^{-\lambda t}$), a repair time MTTR (repairable unavailability), or a test interval T (periodically tested standby, $\lambda T/2$).
- A **house event** is a switch set to TRUE or FALSE.
- Common-cause groups** apply the β -factor model to a set of basic events.

Calculating

Press **Calculate**. The top-event probability appears above the tree, and basic events are shaded by their Fussell-Vesely importance. The **Results** tab gives:

- the exact top-event probability (binary decision diagram), the rare-event approximation and the min-cut upper bound;
- single points of failure;
- minimal cut sets with probability and share;
- Birnbaum and Fussell-Vesely importance for every basic event.

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FTA

FTA — loss of surveillance picture (Manual §14.3)

2 participants

Save

Fault treeResults

Top-event probability (exact, BDD)	1.2200e-6
Rare-event approximation	1.2200e-6
Min-cut upper bound	1.2200e-6
Single points of failure	E4
Engine	fta-1.1.0 · 31 BDD nodes

Minimal cut sets

Cut set	Order	Probability	Share
{E4}	1	1.00e-6	82.0%
{E5, E6}	2	2.00e-7	16.4%
{E7, E8, E9}	3	2.00e-8	1.6%
{E1, E2, E3}	3	5.00e-13	0.0%

Importance measures

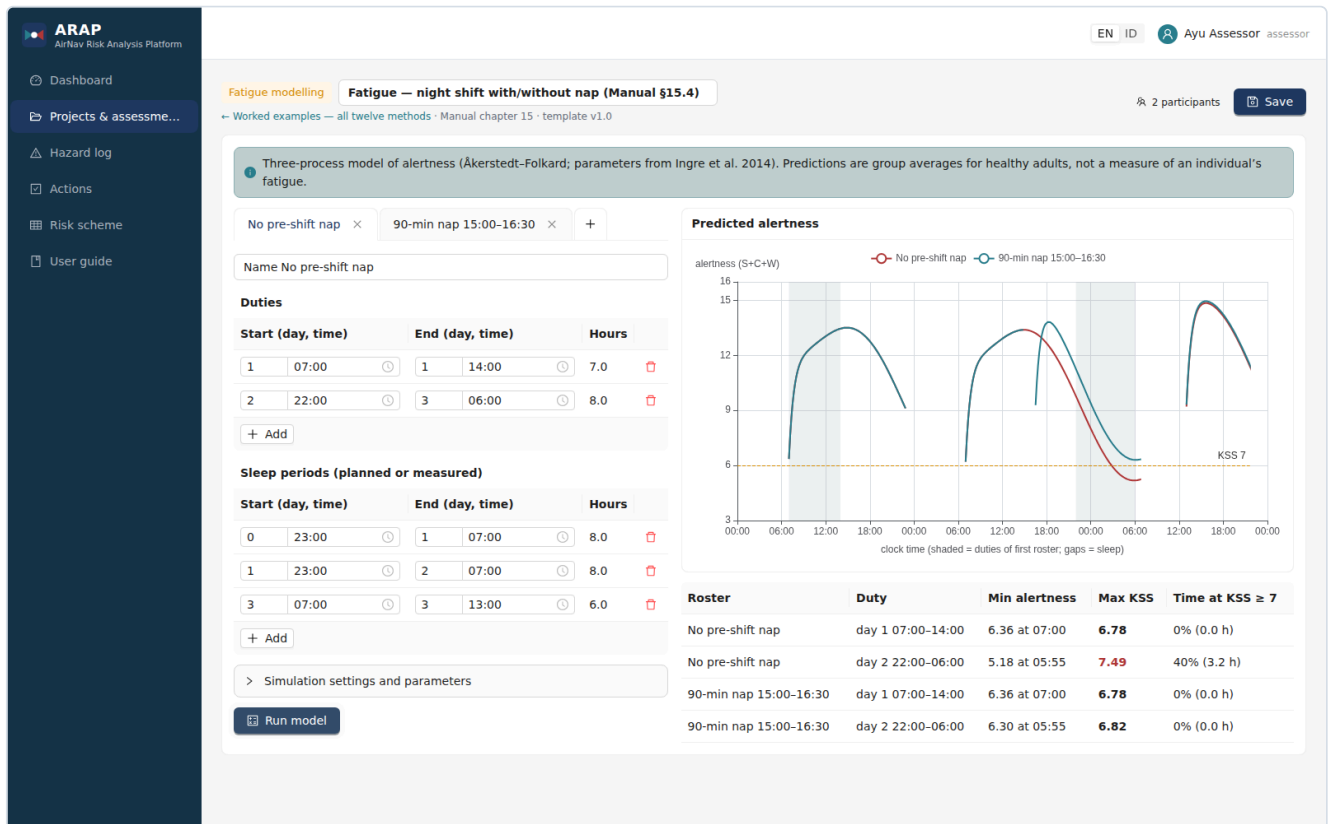
Event	Description	p	Birnbaum	Fussell-Vesely
E4	Common-cause loss of surveillance WAN	1.00e-6	1.00	0.820
E5	Main surveillance data processor fails	2.00e-5	1.00e-2	0.164
E6	Fallback fails to take over (PFD)	0.0100	2.00e-5	0.164
E7	Mains power failure	1.00e-3	2.00e-5	0.0164
E8	UPS fails	2.00e-3	1.00e-5	0.0164
E9	Generator fails to start (PFD)	0.0100	2.00e-6	0.0164
E1	Radar A fails	1.00e-4	5.00e-9	4.098e-7
E2	Radar B fails	1.00e-4	5.00e-9	4.098e-7
E3	ADS-B ground network fails	5.00e-5	1.00e-8	4.098e-7

FTA results

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9. Fatigue modelling



Fatigue model

1. Each tab is a **roster variant** (e.g. *No pre-shift nap* and *90-min nap*). Use + to copy the first roster into a new variant.
2. Enter **duties** and **sleep periods** as day number and clock time. Day 1 starts at 00:00; a sleep starting at 23:00 the evening before day 1 is entered as day 0, 23:00.
3. **Simulation settings and parameters** sets the number of days, time step, KSS threshold and, if needed, the model parameters (leave blank for the defaults of the Manual, §15.2).
4. Press **Run model**. The chart shows predicted alertness for each variant (gaps are sleep; shaded bands are duties of the first variant; the dashed line is the KSS threshold). The table gives, per duty, the minimum alertness and when it occurs, the maximum KSS, and the time spent at or above the threshold.

Predictions are group averages for healthy adults. They are not a measure of an individual's fatigue and must be checked against operational data (Manual §15.6).

Samn-Perelli fatigue check. The second tab of the fatigue study records how each controller feels before taking over a position (Manual §15.5). Click + **Rating** and enter the date, time, shift, controller ID (not the name), planned position and the self-rated score:

Score	Description	Light and required action
1	Fully alert, wide awake	Green (1-3): proceed to the shift normally
2	Very lively, responsive, but not at peak	
3	Okay, somewhat fresh	
4	A little tired, less than fresh	Caution: standard position (no highly complex, high-traffic sectors) and a slightly earlier break
5	Moderately tired, let down	Amber: immediate mitigation before taking a position — e.g. a mandatory 15-minute walk, caffeine, or a rotation change
6	Extremely tired, very difficult to concentrate	Red (6-7): operational risk — remove from active control (administrative desk tasks) or send home to rest
7	Completely exhausted, unable to function	

NAVRAP shows the light for each score. Record the supervisor's response and, for amber, the mitigation applied. A response weaker than the threshold requires is flagged (red for a red-light controller left on a position; orange otherwise), and so is a non-green score with no response. The summary shows the count per light, the score distribution and the share of non-green ratings; the Safety Assessment Report includes it.

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Fatigue modelling

Fatigue — night shift with/without nap (Manual §15.4)

2 participants

Save

Three-process model (rotor)

Samn-Perelli fatigue check 6 1 to act on

1 Samn-Perelli fatigue checklist

A 7-point self-rating of mental exhaustion and cognitive slowing, taken before the controller takes over a position. This three-process model predicts group-average alertness for a rotor; this rating records how the individual feels now. Record controller IDs, not names — ratings are personal data and are used to manage fatigue, not to judge performance.

Scale and action thresholds

1 Fully alert, wide awake

2 Very lively, responsive, but not at peak

3 Okay, somewhat fresh

4 A little tired, less than fresh

5 Moderately tired, let down

6 Extremely tired, very difficult to concentrate

7 Completely exhausted, unable to function

Scores 1-3: Green light. The controller can proceed to their shift normally.

Score 4: Caution light. The supervisor assigns the controller to a standard position (avoiding highly complex, high-traffic sectors) and schedules their break slightly earlier.

Score 5: Amber light. Immediate mitigation before the controller takes over a position — for example a mandatory 15-minute walk, a dose of caffeine, or adjusting the rotation schedule.

Scores 6-7: Red light. The controller is an operational risk: pull them from active control duties, place them on administrative desk tasks, or send them home to rest.

Summary

6 ratings(s)

Green light: 2 Caution light: 2 Amber light: 1 Red light: 1

mean 4.0 - 67% not green

1 response check(s)

ATC-09 score 4 (Caution light) but recorded 'Proceeded to the shift normally' — The supervisor assigns the controller to a standard position (avoiding highly complex, high-traffic sectors) and schedules their break slightly earlier.

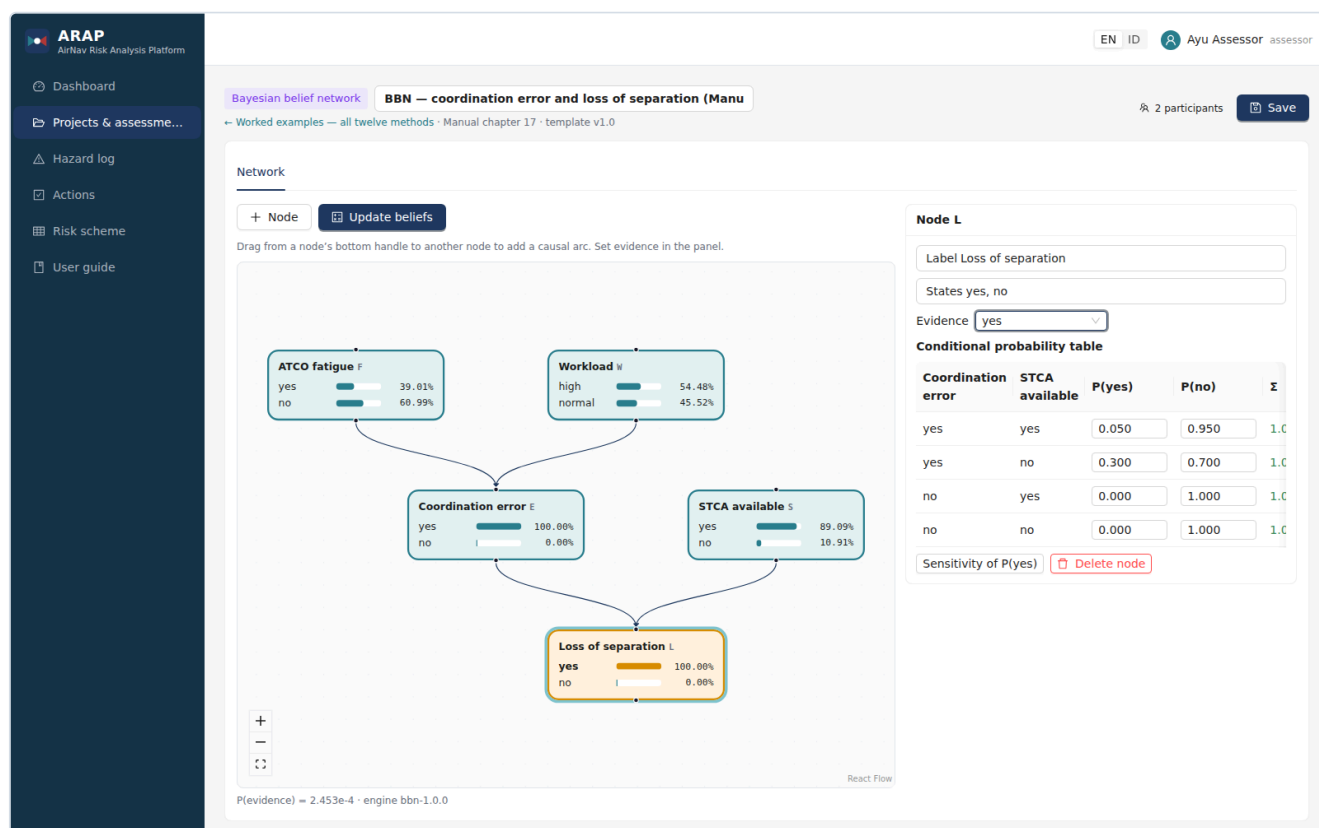
Pre-shift ratings

Date	Time	Shift	Controller ID	Planned position	Score	Light / required action	Supervisor's response
2026-09-	21:45	Night	ATC-07	APP West	2 Very lively, responsive, but not at peak	Green light	Proceeded to the shift normally
2026-09-	21:45	Night	ATC-12	APP East	3 Okay, somewhat fresh	Green light	Proceeded to the shift normally
2026-09-	21:50	Night	ATC-03	TWR	4 A little tired, less than fresh	Caution light	Standard position, earlier break
2026-09-	21:50	Night	ATC-21	ACC Sector	5 Moderately tired, let down	Amber light	Mitigated before taking a position
2026-09-	21:55	Night	ATC-15	ACC Sector	6 Extremely tired, very difficult to concentrate	Red light	Removed from control — sent home
2026-09-	05:40	Morning	ATC-09	APP West	4 A little tired, less than fresh	Caution light	Proceeded to the shift normally

+ Rating

Samn-Perelli fatigue check

10. Bayesian belief networks



Bayesian network with evidence

Building the network

- **+ Node** adds a node; drag nodes to arrange them.
- To add a causal arc, drag from the **bottom handle** of the cause to the effect node. Double-click an arc to remove it. NAVRAP rejects cycles when calculating.
- Select a node to edit its label and **states** (comma-separated) and its **conditional probability table** — one row per combination of parent states. Each row must sum to 1; the Σ column turns red when it does not (for two-state nodes the second value fills in automatically).

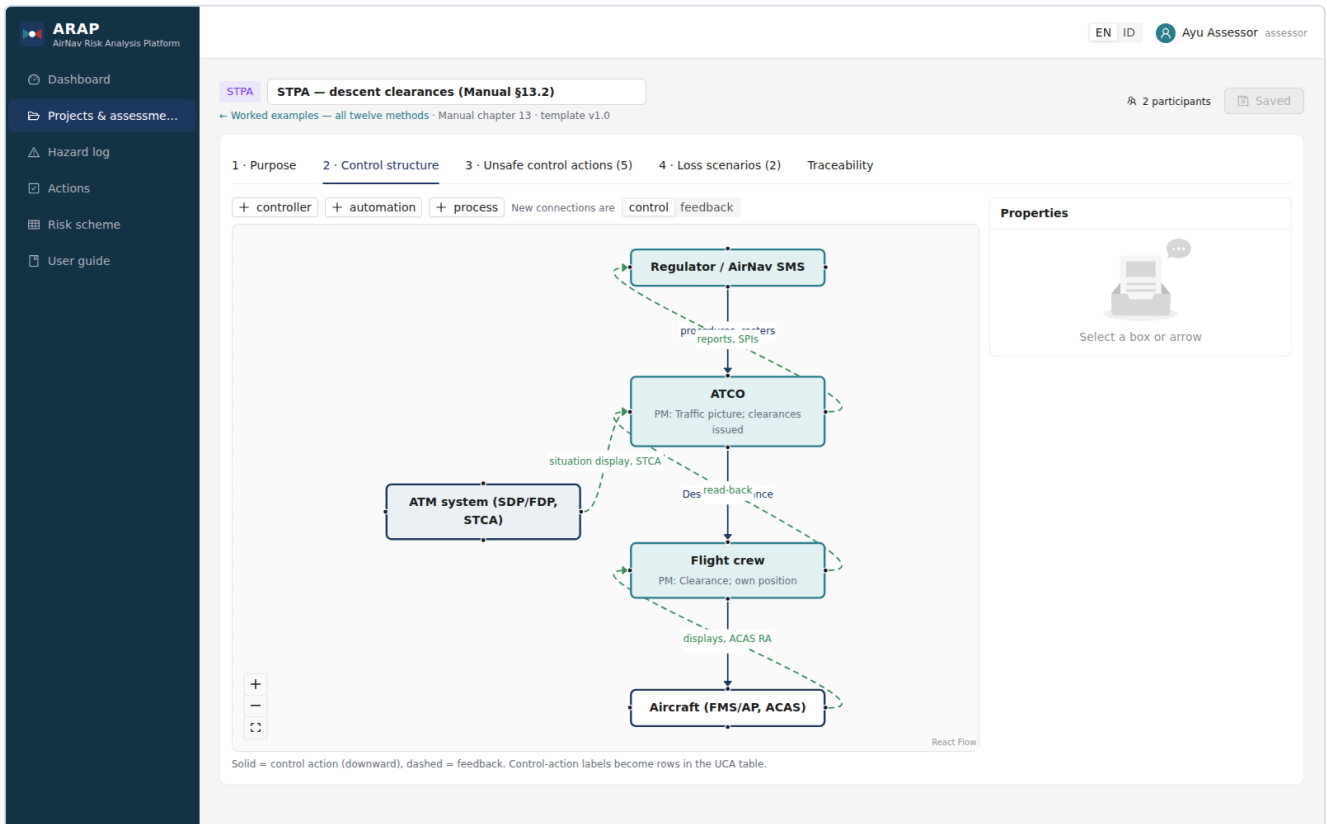
Using the network

- **Update beliefs** calculates every node's probabilities; the bars inside the nodes show them.
- Select a node and choose a state under **Evidence** to condition on it (the node turns amber). All other nodes update — this is how you ask diagnostic questions such as *given a loss of separation, how likely was fatigue?*
- **Sensitivity** shows how much each other node can move the selected node's probability (tornado chart).

11. STPA

The STPA study follows the four steps of the STPA Handbook, one tab each.

1. **Purpose** — losses, system-level hazards (linked to losses) and system-level constraints (linked to hazards).
2. **Control structure** — add controllers, automation and controlled processes; drag from a box's bottom/right handle to another box to add an arrow. Choose whether new arrows are **control** actions (solid) or **feedback** (dashed). Click a box to write its *process model*; click an arrow to rename it.



STPA control structure

3. **Unsafe control actions** — for every control action, four columns (*not providing, providing, too early/too late/wrong order, stopped too soon/applied too long*). Add UCAs with **+ UCA**; each needs a context ("when ...") and at least one hazard, otherwise NAVRAP warns.

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STPA

STPA — descent clearances (Manual §13.2)

Worked examples

all twelve methods

Manual chapter 13

template v1.0

2 participants

Saved

1 Purpose

2 Control structure

3 Unsafe control actions (5)

4 Loss scenarios (2)

Traceability

Control action: procedures, rosters

Not providing causes hazard

Providing causes hazard

Too early / too late / wrong order

Stopped too soon / applied too long

Control action: Descent clearance

Not providing causes hazard

UCA-1

ATCO does not issue descent

when the aircraft must leave its level to avoid a conflict ahead

H-1

remove

+ UCA

Providing causes hazard

UCA-2

ATCO clears descent through an occupied level

when unseparated traffic is at that level

H-1

remove

UCA-3

ATCO clears descent below MVA

when vectoring over high terrain

H-2

remove

+ UCA

Too early / too late / wrong order

UCA-5

Descent issued before preceding aircraft vacated level

when relying on the label trend

H-1

remove

+ UCA

Stopped too soon / applied too long

+ UCA

Control action: Execute descent

Not providing causes hazard

Providing causes hazard

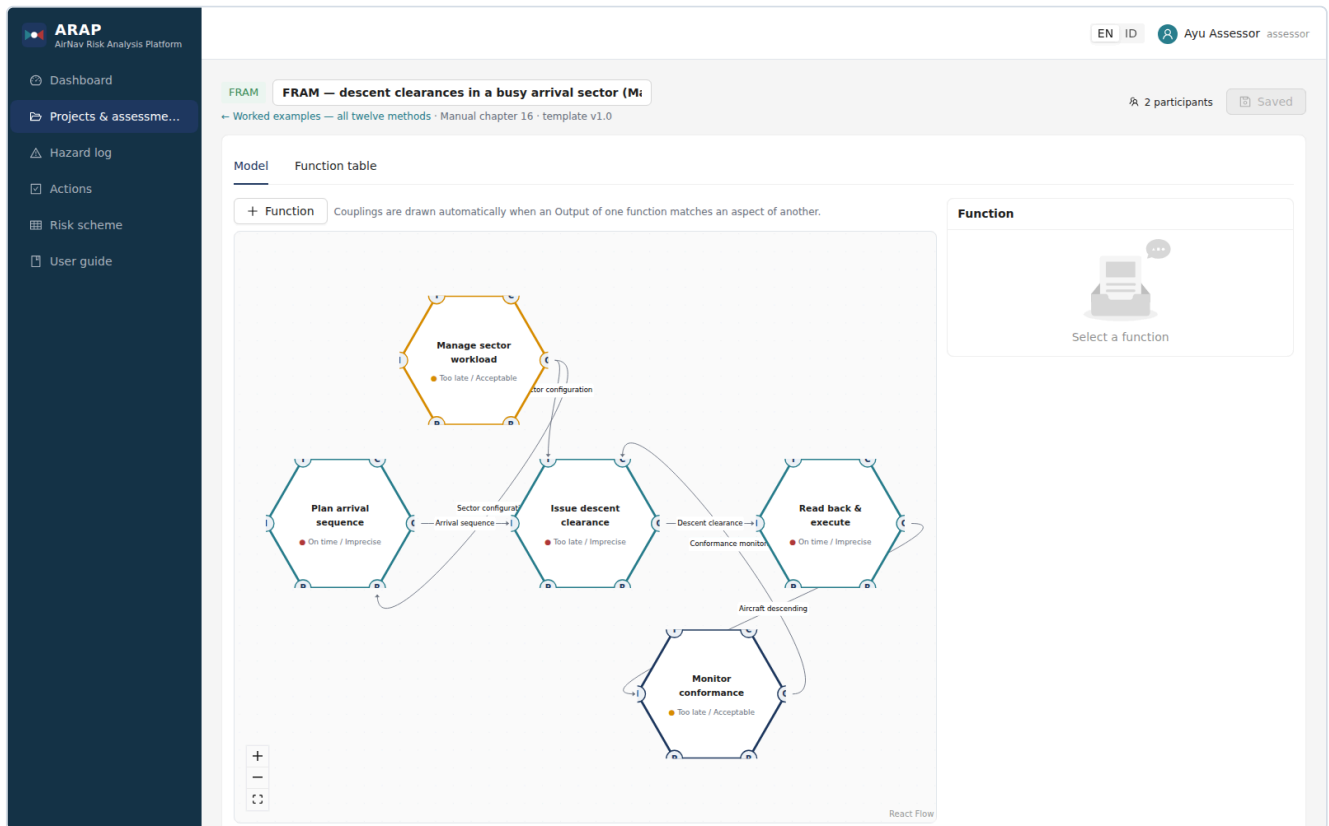
Too early / too late / wrong order

Stopped too soon / applied too long

UCA table

- Loss scenarios** — for each UCA, the scenario type, description and the resulting requirement or constraint. **Send hazards to hazard log** creates one hazard-log entry per STPA hazard.
- Traceability** — the chain loss → hazard → constraint/UCA → scenario → requirement, with broken chains flagged.

12. FRAM



FRAM model

- **+ Function** adds a function (hexagon). Drag to arrange.
- Select a function to set its name, type (human, technological, organisational) and the six **aspects** — Input, Output, Precondition, Resource, Control, Time. Type an entry and press Enter; existing outputs are offered as suggestions.
- **Couplings are drawn automatically** wherever an Output of one function has the same text as an aspect of another. Unmatched aspects are listed so you can fix typos or add missing functions.
- Record the **output variability** (timing: too early / on time / too late / not at all; precision: precise / acceptable / imprecise). The dot colour on each hexagon shows it.
- The **Function table** tab lists everything in one table.

13. Extended methods (Manual chapters 18–31)

Version 0.2 adds thirteen methods, and version 0.3 adds the wildlife strike module. They all work like the others: open the study, edit, press **Calculate** where there is one, and **Save**. The demo project **DEMO-02** has one worked example of each, matching the Manual.

13.1 Collision risk — CRM (Manual §18)

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AirNav Risk Analysis Platform

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Collision risk modelling **CRM — RVSM vertical risk and parallel-route spacing**

Worked examples — thirteen extended methods · Manual chapter 18 · template v1.0

2 participants Save

Reich collision risk model in occupancy form (ICAO Doc 9689 / Doc 9574). Overlap probabilities and occupancies should come from regional monitoring agency data (e.g. height-keeping and lateral deviation monitoring).

Vertical (RVSM) Lateral (route spacing)

Parameters

λ_x length (ft) 184 λ_y wingspan (ft) 175

λ_z height (ft) 50

Lateral deviation model Double expone... scale (NM) 0.9

Route spacing S_y (NM) 15

$P_z(0)$ — vertical overlap, same level 0.52 $E_y(\text{same})$ — same-direction lateral occupancy 0.1

$E_y(\text{opp})$ — opposite-direction lateral occupancy 0 S_x — longitudinal window (NM) 80

$|\Delta v|$ (kt) 20 $|v|$ (kt) 470

$|z|$ (kt) 1.5 $|y(S_y)|$ lateral closing speed at loss of spacing (kt) 4

Target level of safety 0.000000005

Calculate

Result

$P_y(S_y)$ at 15 NM	1.63e-8
Lateral risk (fatal accidents / fh)	1.578e-10 Meets TLS
Minimum spacing meeting TLS	11.68 NM

collision risk / fh

min 11.7 NM

route spacing (NM)

CRM lateral spacing

Two tabs. **Vertical (RVSM)** takes the Reich-model parameters: overlap probabilities, occupancies, speeds and aircraft dimensions (entered in feet). **Calculate** shows the same- and opposite-direction risk, the total and whether it meets the target level of safety (TLS). **Lateral (route spacing)** also asks for the lateral-deviation model (double exponential or Gaussian) and its scale. It plots risk against route spacing on a log scale and gives the **minimum spacing** that meets the TLS. The demo gives 1.87×10^{-9} (vertical) and 11.7 NM (lateral).

13.2 Event tree analysis — ETA (Manual §19)

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Event tree analysis

ETA — conflict after a level bust (Manual §19.3)

Worked examples — thirteen extended methods · Manual chapter 19 · template v1.0

2 participants

Save

Initiating event

Conflict following a level bust

Frequency 0.025 per year

A successful barrier ends the sequence (barrier event tree)

Barriers / functional events (in order)

ID	Barrier	P(success)
B1	ATCO detects and resolves	0.90
B2	STCA alert and ATCO resolution	0.90
B3	ACAS II RA followed	0.90
+ Add		

Conditional probabilities (depend... path prefix e.g. "F" or "F-F")

After path	Barrier	P(success)
F	B2	0.80
+ Add		

Calculate

Initiating event

Conflict following a level bust

B1

ATCO detects and resolve

B2

STCA alert and ATCO reso

B3

ACAS II RA followed

D

Resolved by ATCO — no loss of separation

S · 0.0225 per year

E

Loss of separation, resolved after STC

F-S · 2.00e-3 per year

F

Serious loss of separation, resolved by ACAS

F-F-S · 4.50e-4 per year

A

Near mid-air collision / collision

F-F-F · 5.00e-5 per year

Up (green) = barrier succeeds · down (red) = barrier fails

Outcomes

Sequence	Outcome	Severity	Probability	Frequency (per year)
S	Resolved by ATCO — no loss of separation	D	0.900	0.0225
F-S	Loss of separation, resolved after STCA	C	0.0800	2.000e-3
F-F-S	Serious loss of separation, resolved by ACAS	B	0.0180	4.500e-4
F-F-F	Near mid-air collision / collision	A	2.000e-3	5.000e-5

Σ probability = 1.000000 · by severity: D 0.0225 · C 2.00e-3 · B 4.50e-4 · A 5.00e-5 · edit outcomes then recalculate

Event tree

1. Enter the initiating event and its frequency.
2. List the barriers **in the order they act**, with their probability of success.
3. Under **Conditional probabilities**, enter any dependence as a path prefix. For example, **F / B2 / 0.8** means "STCA succeeds 80% of the time when the controller has already failed".
4. Press **Calculate**. NAVRAP draws the tree and lists each sequence with its probability and frequency.
5. Name each outcome and give it a severity, then recalculate. The totals by severity appear under the table.

13.3 Human reliability — HRA (Manual §20)

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Human reliability (HEART / CARA)

HRA — controller and technician tasks (Manual §20.4)

2 participants Save

HEP = $GTT \times \prod[(EPC - 1) \times APOA + 1]$. Use CARA for controller tasks and HEART for other tasks (e.g. engineering). Justify every APOA; use the HEP in FTA, LOPA or ETA.

Tasks

ID	Task	GTT	HEP
T1	ATCO acts correctly on an STCA alert	CARA B3	2.016e-3
T2	ATCO detects a level bust by monitoring the display (no alert)	CARA B1	0.0575
T3	Technician restores radar to operational configuration after maintenance	HEART F	0.0108

Calculate all (server)

Task T1

ATCO acts correctly on an STCA alert

CARA (ATC)

HEART

B3 — Respond to unique and trusted audible and visual i...

Error-producing conditions

EPC	Condition	Max	APOA	Factor
17	Traffic complexity	×10		2.80
6	Cognitive overload	×6		1.50
13	Low vigilance	×3		1.20

Add an error-producing condition

Nominal HEP 0.0004 → assessed HEP 2.016e-3

Justification of the task type and APOA values

HRA

Add a task and choose **CARA** (controller tasks) or **HEART** (other tasks) and the generic task type. Add error-producing conditions from the list. For each one, set the **APOA** slider (0–1) and write a justification. The HEP updates as you edit. **Calculate all** runs the server engine for the record.

13.4 Occurrence risk classification — ERC and RAT (Manual §21)

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Occurrence risk classification
Occurrence risk classification — ERC and RAT (Manual §
2 participants
Saved

Occurrences

ID

Occurrence

ERC

RAT

Class

OCC-101	Level bust FL240→FL23...	102	7	B
OCC-102	Vehicle entered runway...	50	—	
OCC-103	Frequency congestion ...	1	—	

Send red-band occurrences to hazard log

OCC-101

Level bust FL240→FL232 with crossing traffic; STCA; ATCO resolved; 2.8 NM / 700 ft

Select date

ESARR 2 severity class (analyst)

B

ARMS ERC

RAT scoring

Minimum separation achieved

Separation ≥ minimum (0)

> 75% of minimum (1)

> 50% and ≤ 75% (3)

> 25% and ≤ 50% (7)

≤ 25% of minimum (10)

Rate of closure

None (0)

Low (≤ 85 kt, ≤ 1000 ft/min) (1)

Medium (≤ 205 kt, ≤ 2000 ft/min) (2)

High (≤ 700 kt, ≤ 4000 ft/min) (4)

Very high (> 700 kt, > 4000 ft/min) (5)

Conflict detection (ATM ground)

Detected (0)

Detected late (1)

Not detected (2)

Not applicable (0)

Planning (ATM ground)

Plan correct (0)

Plan inadequate (1)

No plan (2)

Not applicable (0)

Execution

Execution correct (0)

Execution inadequate (1)

No execution (2)

Not applicable (0)

Ground safety net (STCA / MSAW / APW)

Triggered (0)

Not triggered (2)

Not applicable (0)

Recovery

Recovery correct (0)

Recovery inadequate (1)

No recovery / worsened (2)

Not applicable (0)

Airborne safety net / pilot initiative

TCAS/GPWS or see-and-avoid effective (0)

No RA / warning (2)

Not applicable (0)

Risk of collision 5 + controllability 2 = severity score 7

Assign the ESARR 2 severity class with the EUROCONTROL RAT look-up table (not embedded in ARAP) and record it above.

Notes / rationale

ERC

- **Add an occurrence.** Add a short description, date and occurrence ID.
- **ARMS ERC.** Click the matrix cell that answers Q1 (most credible accident outcome) and Q2 (effectiveness of the remaining barriers). The band and recommended response are shown.
- **RAT scoring.** Pick one answer per item. NAVRAP adds the points for risk of collision and controllability. Record the ESARR 2 severity class that you read from the EUROCONTROL RAT table.
- **Send red-band occurrences to hazard log.** Creates hazard entries for the red-band occurrences.

13.5 Safety argument — GSN (Manual §22)

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EN ID Ayu Assessor assessor

Safety case (GSN) **Safety argument — ADS-B introduction (Manual §22.3)** 2 participants Saved

← Worked examples — thirteen extended methods · Manual chapter 22 · template v1.0

Structure valid 6 goals 4/4 solutions linked to evidence 1 undeveloped

Solid arrow = SupportedBy · open arrow, dashed = InContextOf · ◊ = undeveloped. Click an element to edit.

GSN Diagram:

- G1** Introduction of ADS-B surveillance in en-route sector II is acceptably safe
 - C1** Scope, environment and assumptions of assessment DEMO-G1
 - G2** Every FIA failure condition meets its safety objective
 - S2** Argue over each failure condition FC-1 ... FC-3
 - G3** FC-3 (undetected erroneous position, one aircraft) $\leq 1 \times 10^{-4}$ per operating hour
 - Sn1** FIA worksheet — objectives (Risk #7 · open)
 - Sn2** FTA result for the integrity-monitoring architecture (Risk #9 · open)
 - G4** FC-4 (undetected erroneous position, several aircraft) $\leq 1 \times 10^{-4}$ per operating hour
 - A1** Radar coverage above FL245 remains available as an independent cross-check (Risk #10 · open)
 - S1** Argue over the specification, the implementation and the transition
 - G5** Residual risks are ALARP and accepted by the proper authority
 - Sn3** Hazard log entry HZ-0003 with acceptance... (Risk #003 · open)
 - J1** Acceptance authorities follow Manual Table 3.3
 - G6** Transition to operations does not increase controller workload beyond acceptable limits
 - Sn4** Real-time simulation results (workload, R/T occupancy) (Risk #22 · open)

Sn4 — solution

Real-time simulation results (workload, R/T occupancy)

Parent G6

Evidence

ARAP study

#22 SIM — Simulation — new arrival procedu...

Status: **SIM #22 · open · open study**

Evidence is not yet complete / accepted — the argument is provisional.

Argument checks

✓ No structural issues found.

GSN

- **Build the argument.** Click an element to edit it. Use **+ goal** / **+ strategy** / **+ solution** / **+ context** / **+ assumption** / **+ justification** to add elements below it. NAVRAP lays out the diagram automatically.
- **Link evidence.** For a **solution**, link the evidence: an NAVRAP study, a hazard log entry or an external document. The solution shows the evidence status in green (complete or accepted) or red (still provisional).
- **Argument checks.** This panel lists structural problems: more than one top goal, unsupported goals, strategies supported by something other than goals, solutions without evidence, and cycles. Mark a goal **undeveloped** when its support is still to come.

13.6 Common cause analysis — CCA (Manual §23)

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EN ID Ayu Assessor assessor

Common cause analysis Common cause analysis — ACC surveillance chain (Man) 2 participants Save

Worked examples — thirteen extended methods · Manual chapter 23 · template v1.0

Every AND gate or redundancy claim in an FTA, RBD or FHA rests on independence. CCA checks it three ways: where items are installed (ZSA), what single events can hit several at once (PRA) and what they share by design (CMA).

3 independence claim(s) not supported 0 claim(s) not yet assessed 3 particular risk(s) defeat redundancy Send failures of independence to hazard log

Zonal safety analysis (3) Particular risks (4) Common mode analysis (3)

+ Add row Delete selected Import claims from FTA

ID	Independence claim	Items claimed indepe...	Common-mode...	Analysis	Indepen...	Action
CM-01	☐ Main SDP (E5) and fallback (E6) fail independently	E5, E6	Software	Fallback runs the same software baseline as the main system	No	Diverse fallback or β -factor in FTA
CM-02	☐ Mains (E7), UPS (E8) and generator (E9) fail independently	E7, E8, E9	Location	UPS and generator switchgear share the power room (fire zone)	Partly	Include fire CCF in FTA
CM-03	☐ Radar A, radar B and ADS-B inputs are independent	E1, E2, E3	Shared utilities (power, HVAC, network)	All three use the single surveillance WAN (E4)	No	Diverse data path (FTA shows 82% of risk)

CCA

Three worksheets: **Zonal safety analysis**, **Particular risks** and **Common mode analysis**. On the common-mode tab, **Import claims from FTA** creates one independence claim for every AND or k-out-of-n gate in a chosen fault tree. For each claim, record the common-mode source and whether independence holds. The tags at the top count unsupported claims and particular risks that defeat redundancy. **Send failures of independence to hazard log** passes them on.

13.7 RBD and Markov (Manual §24)

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RBD and Markov analysis **RBD and Markov — sector VHF availability (Manual §24).** 2 participants Save

← Worked examples — thirteen extended methods · Manual chapter 24 · template v1.0

Reliability block diagram Markov model

series: Sector frequency available at CWP

parallel: Radio chains

Main Tx/Rx chain
MTBF 20000 h · MTTR 4 h
A = 0.999800

Standby Tx/Rx chain
MTBF 20000 h · MTTR 4 h
A = 0.999800

parallel: Links to remote site

Leased line
MTBF 4000 h · MTTR 8 h
A = 0.998004

VSAT backup
MTBF 3000 h · MTTR 12 h
A = 0.996016

VCCS
MTBF 50000 h · MTTR 2 h
A = 0.999960

parallel: Power

Mains
MTBF 2000 h · MTTR 2 h
A = 0.999901

UPS
MTBF 30000 h · MTTR 8 h
A = 0.999733

Calculate availability Blocks are assumed independent — check with common cause analysis.

Availability	0.99995174
Unavailability	4.826e-5
Expected downtime	0.423 h/yr (25.4 min)

downtime saved if perfect (min/yr)

VCCS
Leased line
VSAT backup
Mains
UPS
Main Tx/Rx chain
Standby Tx/Rx chain

Selected element

Label Sector frequency available at CWP

series parallel koon block

Add inside: + block + series + parallel + koon

RBD

- **RBD tab.** Click a group or block to edit it. Groups can be **series**, **parallel** or **k-out-of-n**. Blocks have MTBF and MTTR. Add elements inside a group with the + buttons.
- **Calculate availability.** Gives availability, unavailability and downtime per year, plus a chart of how many minutes a year each block would save if it were perfect.
- **Markov model tab.** Define states (tick *Service up?*) and transitions with rates per hour, then press **Solve**. You get steady-state probabilities, availability, failure frequency, system MTBF, mean down time and mean time to first failure.

ARAP

AirNav Risk Analysis Platform

Dashboard

Projects & assessme...

Hazard log

Actions

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User guide

EN ID

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RBD and Markov analysis

RBD and Markov — sector VHF availability (Manual §24.)

Worked examples — thirteen extended methods · Manual chapter 24 · template v1.0

2 participants

Save

Reliability block diagram

Markov model

Use a Markov model where blocks are not independent: shared repair crews, cold/warm standby, imperfect switchover (coverage). Rates are per hour ($\lambda = 1/\text{MTBF}$, $\mu = 1/\text{MTTR}$).

States

ID	Description	Service up?
S2	Main in service, standby ready	☒
S1	On standby, main in repair	☒
S0	No radio chain in service	☐
+ Add		

Transitions

From	To	Rate (/h)	Meaning
S2	S1	0.000049504	main fails, switchover OK
S2	S0	0.000000504	main fails, switchover fails
S1	S0	0.00005	standby fails
S1	S2	0.25	main repaired
S0	S1	0.25	one chain restored
+ Add			

Initial state (for MTTF)

Solve

Availability	0.999997960
Unavailability	2.0396e-6 (1.07 min/yr)
Failure frequency	5.099e-7 /h
System MTBF	1,961,176 h
Mean down time	4.00 h
MTTFF	1,961,565 h (223.9 yr)

State	Description	π (steady state)
S2	Main in service, standby ready	1.00
S1	On standby, main in repair	1.9996e-4
S0	No radio chain in service	2.0396e-6

Markov

13.8 SWIFT (Manual §25)

A worksheet like HAZID. Each row is a *what if...?* question under a prompt category, with consequence, safeguards, severity, likelihood and recommendation. **Prompt category coverage** shows which categories have been considered. Use it to record "considered — nothing credible" for the rest. Rows with a consequence can be sent to the hazard log.

13.9 Hierarchical task analysis — HTA (Manual §26)

The screenshot shows the ARAP (AirNav Risk Analysis Platform) interface. The sidebar on the left contains navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme, and User guide. The main content area is titled 'Hierarchical task analysis' and 'HTA — respond to an STCA alert (Manual §26.3)'. It shows a linked HRA study '#16 HRA — controller and technician tasks (M...)' with 14 tasks and 6 error modes. The task tree is displayed in the center, showing a hierarchy of tasks and sub-tasks. Task 0 is 'Respond to an STCA alert'. The right panel shows details for Task 0, including its description, error modes, and consequences. The bottom right corner of the diagram area says 'React Flow'.

HTA

- **Build the hierarchy.** Select a task and press **Sub-task** to add a step below it. Numbering is automatic (1, 1.1, 1.2 ...).
- **Add plans.** Write a **plan** for every task that has sub-tasks. Tasks without one are flagged in red.
- **Record error modes.** Add error modes to bottom-level tasks.
- **Link to HRA.** Choose a **linked HRA study** at the top, then link a task to one of its HRA tasks to show the HEP.
- **Change the view.** The **Outline** tab shows the same content as an indented table.

13.10 Simulation (Manual §27)

ARAP

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Dashboard

Projects & assessments

Hazard log

Actions

Risk scheme

User guide

Simulation (RTS / FTS)

Simulation — new arrival procedure (Manual §27.4)

2 participants

Save

Worked examples — thirteen extended methods · Manual chapter 27 · template v1.0

Exercise plan

Title RTS of new arrival procedure (sector APP-N)

Real-time simulation

Safety objectives / hypotheses

Show the new procedure does not increase workload or R/T occupancy and does not add conflicts.

Scenarios, traffic samples, runs and design

Peak traffic 38 arrivals/h; 8 runs per condition; 4 ATCO teams; counterbalanced order

Participants (ATCOs, pseudo-pilots)

8 current APP ATCOs, 4 pseudo-pilots

Limitations and validity (realism, learning effects)

Measures and run data

one value per run, comma-separated

ID	Measure	Unit	Better	Criterion	Value	Baseline
M1	Workload (ISA, 1-5)	ISA	lower	threshold	3.5	3.4, 3
M2	STCA alerts per hour	/h	lower	no worse than baseline (+margin)	0	1.2, 0
M3	R/T occupancy	%	lower	threshold	60	58, 63

+ Add

Analyse

Results

All success criteria met.

Measure	Baseline mean ± sd (n)	Solution mean ± sd (n)	Solution 95% CI	Δ	Welch's t-test
M1 Workload (ISA, 1-5), executive ATCO	3.69 ± 0.20 (8)	3.16 ± 0.17 (8)	3.02 – 3.30	-0.52	<0.0
M2 STCA alerts per hour	1.10 ± 0.23 (8)	0.85 ± 0.16 (8)	0.72 – 0.98	-0.25	0.02
M3 R/T occupancy	61.63 ± 2.67 (8)	51.75 ± 2.12 (8)	49.98 – 53.52	-9.88	<0.0

% of baseline mean

Baseline Solution Solution 95% CI

Simulation

- **Plan the exercise.** Record the exercise plan: objectives, scenarios, participants and limitations.
- **Define the measures.** For each measure, set whether lower or higher is better and a **criterion**. Use *threshold* (e.g. ≤ 3.5), or *no worse than baseline*, which takes an optional margin. Set the criteria **before** entering results.
- **Enter the run results.** Enter them as comma-separated numbers for the baseline and the solution.
- **Analyse.** You get the mean, standard deviation, 95% confidence interval, Welch's t-test and whether each criterion is met. You can send failed criteria to the hazard log.

13.11 Expert judgement (Manual §28)

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EN ID

Ayu Assessor assessor

Dashboard

Projects & assessme...

Hazard log

Actions

Risk scheme

User guide

Structured expert judgement

Expert judgement — ATCO detection probability (Manu...

2 participants

Saved

Classical model (Cooke)

Delphi

1

Cooke's classical model: each expert gives 5/50/95% quantiles for seed questions (true value known to the facilitator) and target questions. Weights = calibration × information; experts below the significance level α get zero weight.

Expert name

+ Expert

+ Question

α 0.00

4 seed · 1 target

Calculate weights

ID	Question	Seed ?	Realisation	Scale	Expert A			Expert B			Expert C		
					5%	50%	95%	5%	50%	95%	5%	50%	95%
S1	Level busts reported in sector X last year	☒	14	linear	8	12	18	5	10	30	20	25	35
S2	STCA alerts per 1000 flights (sector X)	☒	2.1	linear	1	2	3	0.5	1.5	5	3	4	6
S3	Median ATCO response time to STCA (s)	☒	9	linear	6	9	14	4	8	20	12	15	20
S4	Read-back errors per 1000 clearances	☒	3.5	linear	2	3.5	6	1	4	10	6	8	12
Q1	P(ATCO does not resolve a level-bust conflict before STCA)	☐	target	log	0.03	0.08	0.2	0.01	0.1	0.4	0.1	0.2	0.4

Expert scores

Expert	Calibration	Information	Bins	Weight
Expert A	0.839	0.771	0/2/2/0	0.814
Expert B	0.596	0.249	0/1/3/0	0.186
Expert C	2.54e-5	0.562	4/0/0/0	0.000

Decision maker (combined distribution)

Item	Performance-weighted 5 / 50 / 95%	Equal-weight 5 / 50 / 95%
S1 seed Level busts reported in sector X last year	5.86 / 11.9 / 26.1	5.66 / 16.3 / 33.6
S2 seed STCA alerts per 1000 flights (sector X)	0.658 / 1.97 / 4.35	0.626 / 2.68 / 5.73

Expert judgement

- Classical model (Cooke).** 1. Add experts and questions. 2. Tick **Seed?** for calibration questions and enter the true value. 3. Enter each expert's 5%, 50% and 95% values. Quantiles that are incomplete or not increasing are flagged. 4. Press **Calculate weights** to get calibration, information and weight per expert, and the combined (decision-maker) distribution for every question.
- Delphi.** Add panellists and rounds and enter the estimates, then press **Summarise rounds** to see the median, quartiles and convergence per round.

13.12 Security risk (Manual §29)

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Dashboard

Projects & assessments...

Hazard log

Actions

Risk scheme

User guide

Security risk assessment

Security risk assessment — surveillance and FDP (Mani

2 participants

Saved

Worked examples — thirteen extended methods · Manual chapter 29 · template v1.0

+ Add row

Delete selected

Send all rows to hazard log

Double-click a cell to edit. Tick rows to select.

I.		Asset	Threat scenario	Threat source	Vulnerability	C	I	A	L...	Security risk	Existing controls
SR-01	☐	ADS-B ground station / SDP input	Injection of false ADS-B targets (spoofing)	External attacker	ADS-B messages unauthenticated	1	5	3	3	15 Very high	Radar cross-check available; multilater
SR-02	☐	Flight data processing (FDP)	Ransomware encrypts FDP servers	External attacker	Flat network between office and ops systems	3	4	5	2	10 High	Backups; antivirus
SR-03	☐	VCCS maintenance access	Compromise of vendor remote-maintenance account	Supplier / third party	Shared vendor credentials	3	4	5	2	10 High	VPN
SR-04	☐	ATM adaptation data	Unauthorised change to sector or safety-net parameters	Insider (accidental)	No two-person rule on adaptation changes	1	5	2	2	10 High	Change log

Security risk = $L \times \max(C, I, A)$. High / very high: SR-01, SR-02, SR-03, SR-04. 4 threat scenario(s) have a safety effect — send those to the safety hazard log (ED-205 / Manual §29.3).

Security risk

A worksheet of threat scenarios: asset, threat, source, vulnerability, C/I/A impact (1–5), likelihood (1–5), controls, treatment and owner. **Security risk** = likelihood × the highest of C, I and A, shown as Low / Medium / High / Very high. Give each scenario a **safety effect** (severity A–E) where it has one. Only those rows are sent to the safety hazard log, so exploit details stay in the security study.

13.13 Occurrence investigation — SOAM, HFACS, Tripod Beta (Manual §30)

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AirNav Risk Analysis Platform

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Occurrence investigation Investigation OCC-102 — works vehicle on runway strip

2 participants Saved

Worked examples — thirteen extended methods · Manual chapter 30 · template v1.0

Occurrence SOAM HFACS Tripod Beta

Organisational factors

- Training (TR)**
Contractor drivers not trained on runway closure procedures
- Change management (CM)**
Works plan changed the closure time without a risk assessment
- Communication (CO)**
Revised closure time not briefed to contractor

Contextual conditions

- Workplace conditions**
Night works; poor signage visibility near K3
- Human performance limitations**
Driver's mental model: runway closed for works
- Physiological and emotional factors**
Driver at end of 12-hour shift

Human involvement

- Driver crossed the lit stop bar believing the runway was closed
- TWR did not challenge the driver's position report

Absent / failed barriers

- Awareness** - failed
Driver RT training and licence
- Restriction** - absent
Follow-me / escort for works vehicles
- Restriction** - failed
Stop bar at holding point K3

Occurrence

OCC-102
At night a works vehicle entered the runway strip of 07L without clearance. RIMCAS alerted; the TWR ATCO instructed an aircraft on 4 NM final to go around. Closes

Barriers that worked

- RIMCAS alert
- ATCO go-around instruction

Barriers link failed barriers to the bowtie they belong to

Barrier	Type	Status	Bowtie barrier
Driver RT training and licence	Awareness	failed	PB3 Driver RT training & licence — Runway incursion
Follow-me / escort for works vehicles	Restriction	absent	PB4 Follow-me / escort — Runway incursion

Contextual conditions

Category	Condition
Workplace conditions	Night works; poor signage visibility near K3
Human performance limitations	Driver's mental model: runway closed for works
Physiological and emotional factors	Driver at end of 12-hour shift

Organisational factors

Factor	Finding
Contractor drivers not	

SOAM

- **Occurrence** — the facts, the counts of failed barriers, HFACS categories and actions, and **Send occurrence to hazard log**.
- **SOAM** — barriers (type and status), human involvement, contextual conditions, organisational factors and safety actions. The SOAM chart updates as you type. Link each failed or absent barrier to the **bowtie barrier** it corresponds to. The bowtie then shows "recorded as failed in N investigation(s)" when you select that barrier.
- **HFACS** — tick categories in the four tiers and record the evidence for each.
- **Tripod Beta** — event trios (agent, object, event) with failed or missing barriers traced to immediate cause, precondition, underlying cause and Basic Risk Factor. A BRF profile summarises them.

13.14 Wildlife strike risk — species matrix (Manual §31)

Species risk matrix rows: likelihood · columns: severity

#	Species	Strikes/yr	Per 10k mvt	Damaging	L × S	Risk	Trend
1	Cattle egret (kuntul kerbau) <i>Bubulcus ibis</i>	12.4	1.90	11.3%	5 × 4 = 20	high	▼ 0.09
2	Large flying fox (kalong besar) <i>Pteropus vampyrus</i>	0.6	0.09	33.3%	2 × 5* = 10	moderate	▲ 0.02
3	Pacific golden plover (cereak keryut) <i>Pluvialis fulva</i>	2.2	0.34	9.1%	3 × 3 = 9	moderate	▼ 0.01
4	Javan pond heron (blekok sawah) <i>Ardeola speciosa</i>	4.4	0.67	4.5%	4 × 2 = 8	moderate	▼ 0.06
5	Barn swallow (layang-layang asia) <i>Hirundo rustica</i>	18.0	2.75	0.0%	5 × 1 = 5	low	▼ 0.23
6	Eurasian tree sparrow (burung gereja) <i>Passer montanus</i>	8.0	1.22	0.0%	4 × 1 = 4	low	▼ 0.06

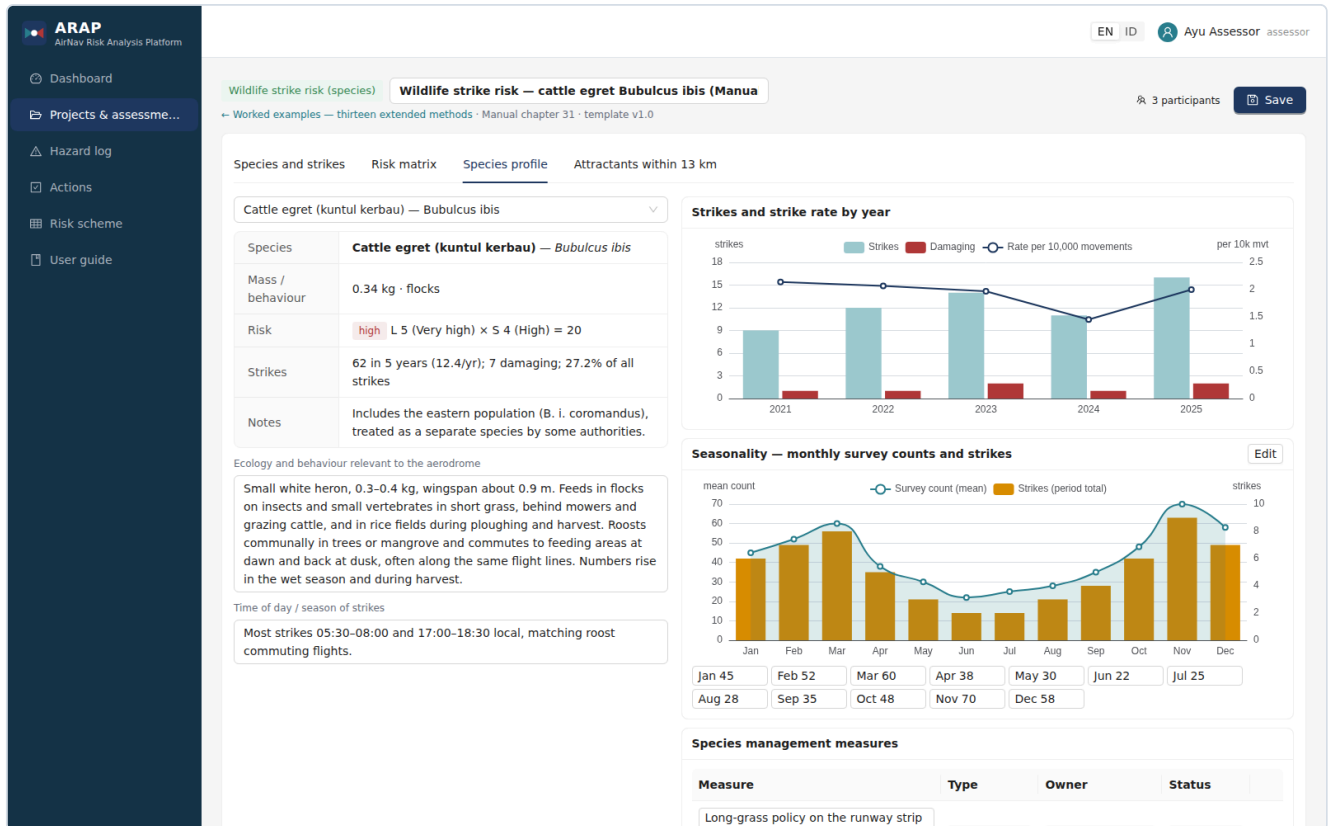
1 high · 3 moderate · 2 low All species: 228 strikes, 6.97 per 10,000 movements
Send high and moderate species to hazard log

* severity from the body-mass surrogate (too few strikes for a stable percentage). Engine wildlife-1.0.0.

Wildlife species risk matrix

- **Species and strikes.** Enter the review years and the aircraft movements for each year. Then add one row per species, or per group of similar species: common and scientific name, body mass, whether it flocks, and the strikes and damaging strikes for each year. Press **Calculate risk**.
- **Risk matrix.** Each species is placed on a 5 × 5 grid. Rows are likelihood (mean strikes per year) and columns are severity (share of strikes that caused damage or had an effect on flight). The ranking table adds the rate per 10,000 movements and its trend. A species with fewer than 5 strikes gets its severity from body mass, plus one class if it flocks; NAVRAP marks this with *. Use **Send high and moderate species to hazard log** to create hazard entries, with the species' attractants and measures.
- **Species profile.** For the selected species: ecology notes, time of day of strikes, a year-by-year chart of strikes and strike rate, a monthly chart of survey counts and strikes (use **Edit** to enter counts), and the species management measures with type, owner and status.
- **Attractants within 13 km.** A register of wildlife attractants: landfills, tambak, rice fields, roosts, grazing land and so on. Each entry has a distance, the species it attracts, an action and an owner. Sites inside the 13 km circle are flagged.

The worked example in **DEMO-02** uses the cattle egret, *Bubulcus ibis*. It has 62 strikes in five years, 7 of them damaging, which gives 5 × 4 = 20 (high). It ranks above the barn swallow, which is struck more often but almost never causes damage.



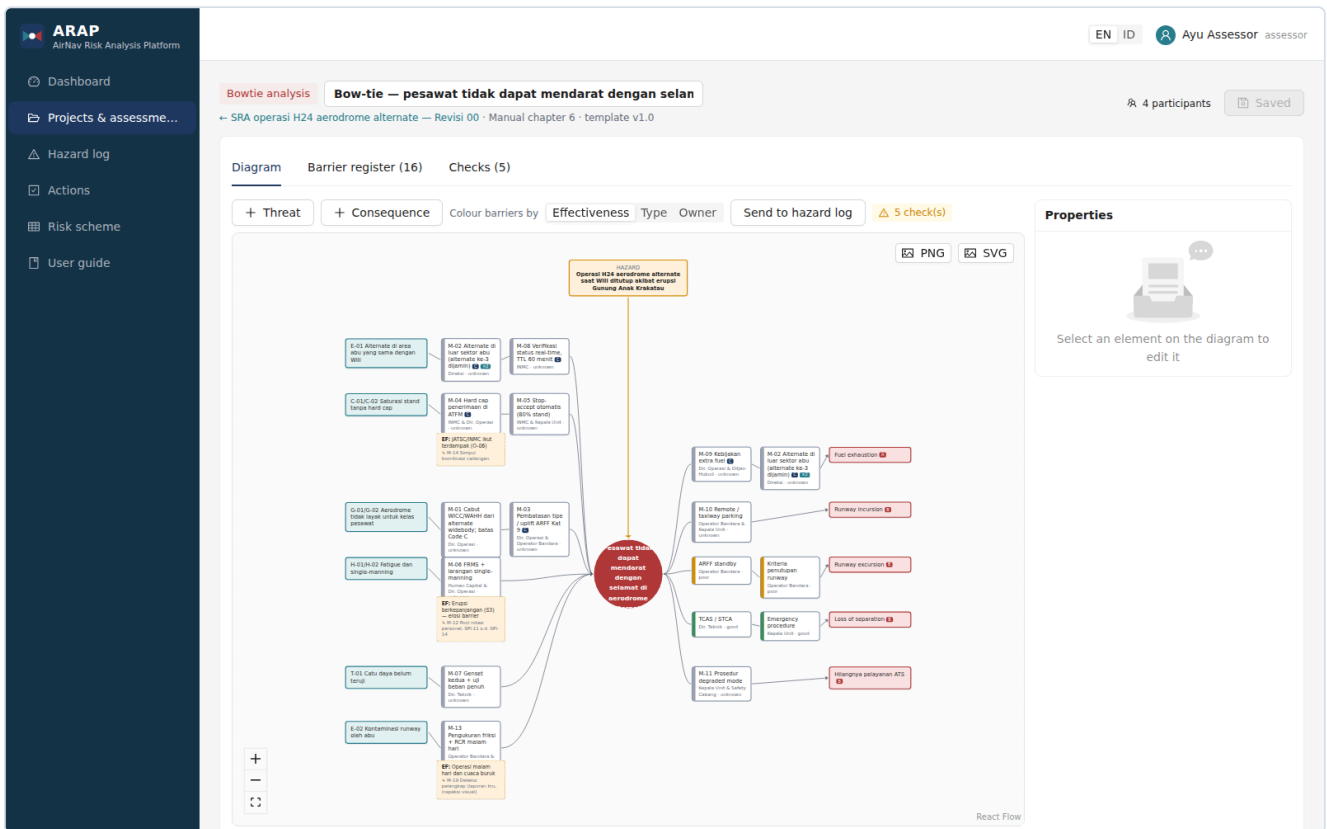
Species profile

13.15 Case study: H24 alternate aerodromes (DEMO-03)

The project **DEMO-03** reproduces a real-format Safety Risk Assessment, *SRA/MOC/OPS/001/IX/2026 Rev 00 — Assessment Operasi 24 Jam Aerodrome Alternate*. It covers activating H24 ATS at seven alternates (WAHI, WAHH, WAHQ, WAHS, WICA, WICC, WIPP) when an eruption of Gunung Anak Krakatau closes WIII. The content is kept in Bahasa Indonesia, as in the source report.

NAVRAP item	What it holds
Assessment scope, environment, assumptions	The change, system boundary, scenarios S1-S3, demand derivation (72-aircraft design wave), feasibility-gate and capacity-model results, and the assumption register A-01 to A-12
HAZID study <i>Register risiko</i>	The 40 hazards from FHA (F1-F7), HAZOP and STPA with their current rating, existing barriers and mitigations
STPA study	The four-organisation closure/reopening decision chain (VAAC → AirNav → operator bandara → KOBU → INMC), six unsafe control actions and their loss scenarios
Two FTA studies	ATS power over 30 days: single untested genset 4.6×10^{-3} , two tested gensets 2.9×10^{-4}
Bowtie	<i>Pesawat tidak dapat mendarat dengan selamat di aerodrome alternate</i> . New barriers M-xx are marked <i>planned</i> , with the SPI that monitors each one
GSN	Claim G0 with sub-claims G1-G7; five are <i>undeveloped</i> because they are waiting for preconditions
Hazard log	ALT-G-01 ... ALT-S-03. The initial rating is the report's <i>Current</i> column, the residual is the <i>Residual</i> column, and the <i>Inherent</i> rating is in the context field. Mitigations M-01 to M-20 are planned controls
Actions	ALT-PK-1 ... ALT-PK-8, the activation preconditions (PK-1, PK-3 and PK-4 are blocking)

The feasibility gate and the capacity (queue/occupancy) model have no dedicated NAVRAP module yet, so their results are recorded in the assessment's environment text.



DEMO-03 bowtie

13.16 Investigation expert system and case library (Manual Appendix E)

The expert system helps in the **initial phase of an ATS occurrence investigation**. It finds similar past occurrences, compares the evidence you have with the evidence they needed, suggests the likely mechanism and — most usefully — lists the evidence worth collecting next. It is advisory: it suggests, investigators conclude, and its outputs do not apportion blame.

It has two parts:

- **Case library** (left menu). Past occurrences stored as AcciMaps in the five ORLIO lanes: Organisational influences, Risk controls, Local conditions, Individual actions, Occurrence events.
- **Investigation expert system (ORLIO)**, a study you add to the assessment of an investigation.

The case library

ARAP

AirNav Risk Analysis Platform

EN ID

Ayu Assessor assessor

Dashboard

Projects & assessments...

Hazard log

Actions

Case library

Risk scheme

User guide

Investigation case library

Past occurrences as ORLIO AcciMaps — the knowledge base of the investigation expert system (Manual Appendix E). Only approved cases are used by search, inference and the Bayesian network.

Draft from report

+ New case

Cases

All categories

All Approved Draft

Ref	Title	Category	Mechanism	Blocks	Source	Status
CASE-01	Similar callsigns — climb clearance taken by the wrong aircraft	ATS — loss of separation	M2 Readback/hearback error not detected	9	manual	approved
CASE-02	Combined sectors at night — converging traffic not detected	ATS — loss of separation	M5 Conflict not detected (monitoring / workload)	10	manual	approved
CASE-03	Trainee issued a level already occupied	ATS — loss of separation	M1 Controller clearance or instruction error	8	manual	approved
CASE-04	Pilot climbed through the cleared level (correct readback)	ATS — loss of separation	M3 Pilot deviation from clearance	7	manual	approved
CASE-05	Aircraft entered the sector without coordination at an FIR boundary	ATS — loss of separation	M4 Coordination failure between units or sectors	8	manual	approved
CASE-06	Conflict inside an STCA inhibition volume	ATS — loss of separation	M6 Safety-net or system deficiency	8	manual	approved
CASE-07	Trainee missed an incorrect readback of heading and level	ATS — loss of separation	M2 Readback/hearback error not detected	9	manual	approved
CASE-08	Weather deviations — conflict between deviating aircraft not detected	ATS — loss of separation	M5 Conflict not detected (monitoring / workload)	9	manual	approved
CASE-09	Level change not coordinated after a sector split	ATS — loss of separation	M4 Coordination failure between units or sectors	8	manual	approved

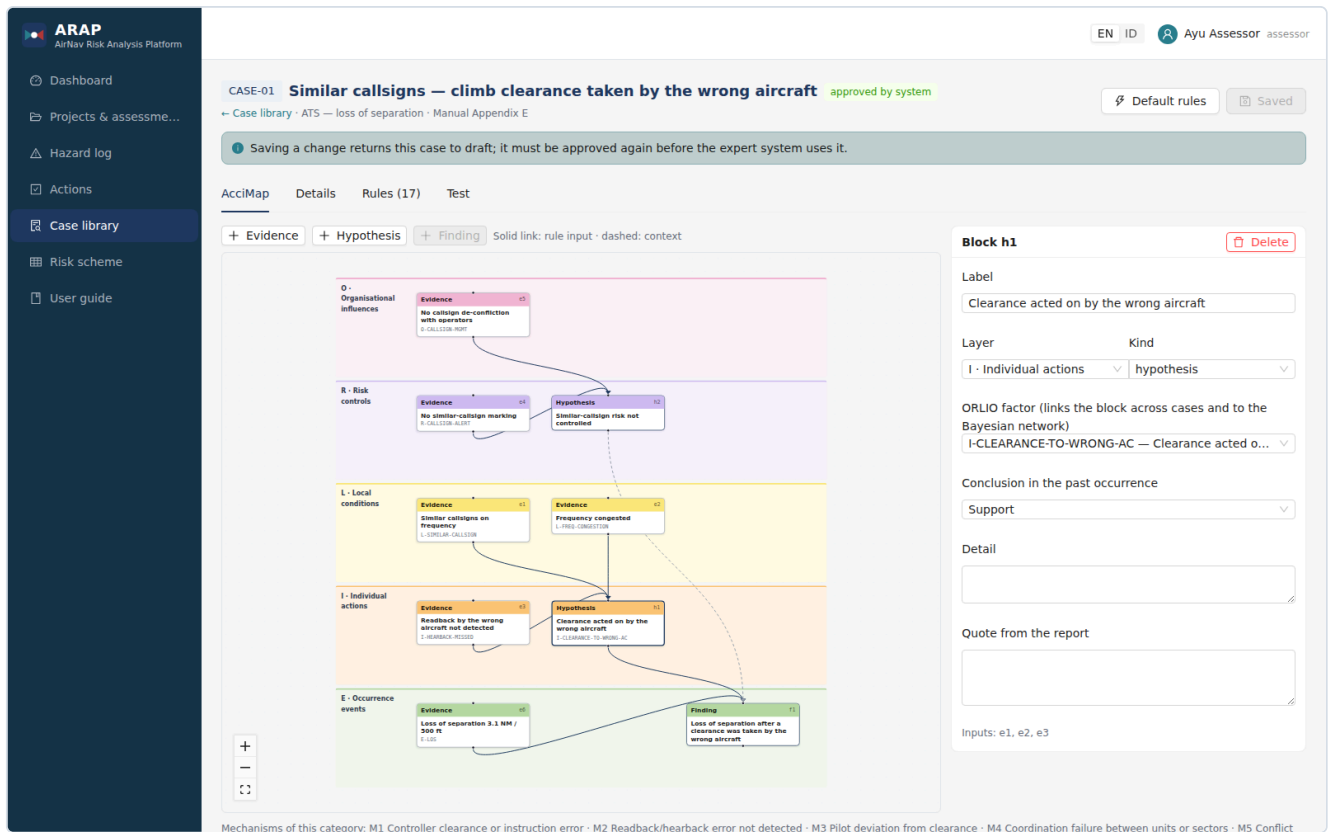
Case library

AcciMaps have a lane for *technical failure mechanisms* beside individual actions, and every catalogue factor shows its ATSB code (§13.17). Verbal probabilities use the ATSB terms: very unlikely, unlikely, about as likely as not, likely, very likely, virtually certain.

Each case has **evidence** blocks (facts, rated in the past occurrence), **hypothesis** blocks (intermediate conclusions) and one **finding** (the occurrence and a verbal probability for each mechanism of the category). Give each block an **ORLIO factor** from the category's catalogue where one fits: the factor links the block across cases and is what the Bayesian network learns from.

In the AcciMap editor:

- **+ Evidence / + Hypothesis / + Finding** add blocks. Click a block to edit it in the panel on the right.
- Drag from a block's lower handle to another block to link them. Click a link to make it an **input** (solid, used by the rules) or **context** (dashed, shown only). Drag a block into another lane to change its layer.
- **Default rules** creates, for each hypothesis and finding, the *confirming* rule (the past states → the past conclusion), a *weakening* rule per input (that input "No effect") and a *contradicting* rule per input (that input opposed). Edit them in the **Rules** tab. A yellow banner warns when the rules no longer match the links.
- The **Test** tab runs the inference with the past states; it should reproduce the past finding.



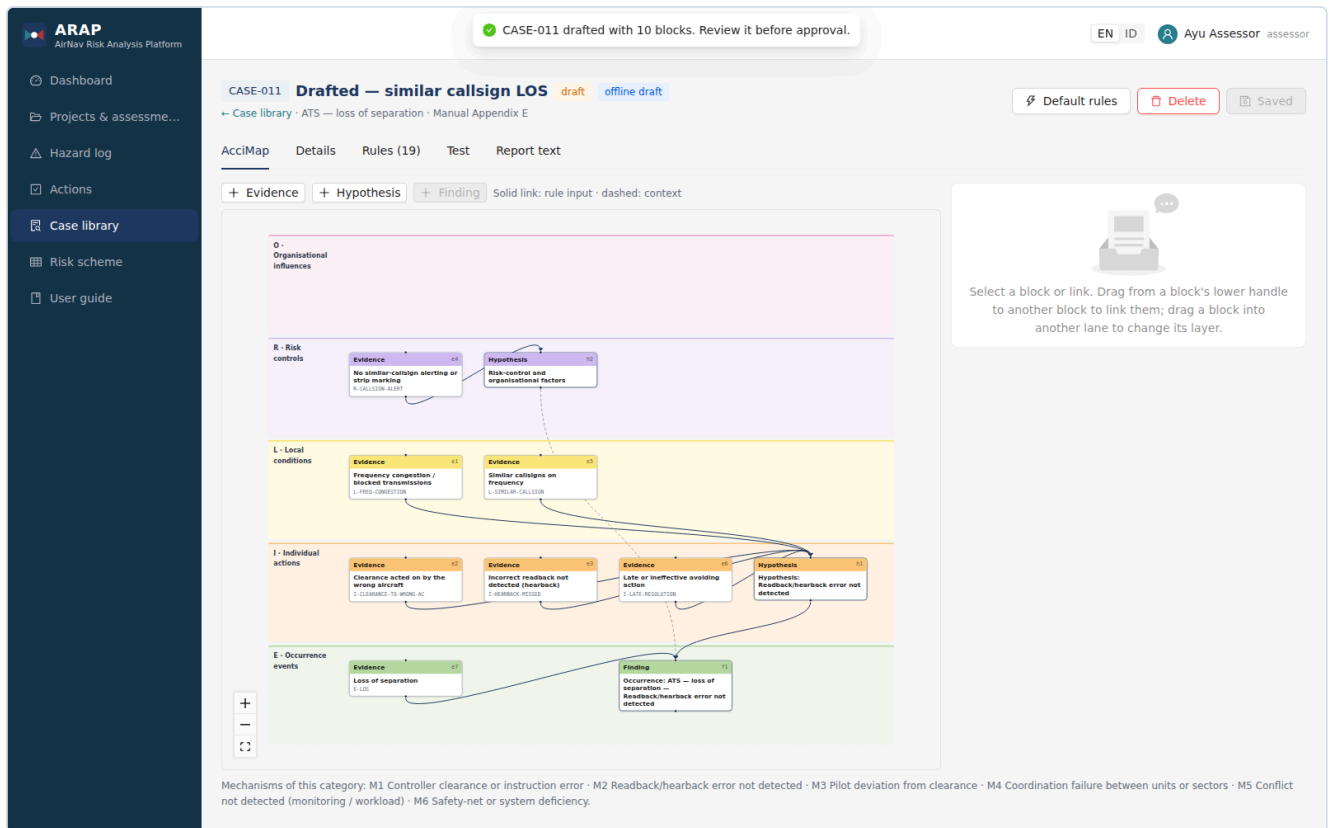
AcciMap editor

A case is used only after a **reviewer approves** it. Saving a change to an approved case returns it to draft.

Draft from report creates a draft AcciMap from pasted text or a text-based PDF of a final report:

- *Offline* (always available): sentences are matched to the factor catalogue and the mechanism keywords.
- *AI (Claude)*: available when the administrator has set `ARAP_ANTHROPIC_API_KEY`. Claude proposes the blocks, links and mechanism. **Only send published final reports**; protected investigation records must not leave AirNav.

Every drafted block carries a quote from the report, which NAVRAP checks against the text. A quote that is not found is marked in red and blocks approval until you correct or clear it; the **Report text** tab lists all quotes. Treat every draft as a starting point: check the layers, the past states and the links, and delete blocks that are not findings of the report.



Draft from a report

Using the expert system in an investigation

Add the study **Investigation expert system (ORLIO)** to the investigation's assessment, choose the occurrence category, and work through the four tabs:

1. **Occurrence & search.** Describe what is known so far and click **Search the case library**. Results show the matched terms and shared factors; tick up to five cases. Factors recognised in your text are shown as suggestions. Click one to rate it *Support*, but check it first.
2. **Evidence ratings.** Rate each factor you have evidence for: *Strongly support* or *Strongly oppose* for recorded or physical evidence (radar replay, RT recording, logs); *Support* or *Oppose* for interviews and judgement; *No effect* when the factor was present but did not matter. Use *Unsure* when evidence exists but its relevance cannot be judged yet. Leave everything else **Not provided**. A factor rated once applies to every case.
3. **AcciMaps.** Click **Run analysis**. Each selected case shows the inferred hypotheses and finding, or *not determined* when evidence is missing. **Clues from the past cases** lists that missing evidence. You can also rate evidence directly in the diagrams.
4. **Bayesian network & ranked clues.** This tab uses the whole approved library of the category. The bar chart shows the prior and posterior probability of each mechanism. The table ranks the unrated factors by **value of information**: how much establishing each would reduce the uncertainty about the mechanism. Rate a clue and run again. **Open as BBN study** creates an ordinary Bayesian-network study with the learned network and your ratings.

ARAP

AirNav Risk Analysis Platform

Dashboard

Projects & assessments

Hazard log

Actions

Case library

Risk scheme

User guide

EN ID

Ayu Assessor assessor

Investigation expert system (ORLIO)

Investigation expert system — OCC-ILL-01

0 participants

Saved

Occurrence category

ATS — loss of separation

3 case(s) selected

7 rating(s)

Run analysis

Bayesian network: **M2 Readback/hearback error not detected** 44% (prior 20%) from 9 approved cases · CASE-01: M2 VP

Advisory only: the system points to similar past occurrences and to evidence worth collecting. Investigators draw the conclusions; findings do not apportion blame or liability. Analysed 2026-09-29 08:01:48+00:00.

1 · Occurrence & search

2 · Evidence ratings (7)

3 · AcciMaps (fuzzy inference)

4 · Bayesian network & ranked clues

CASE-01 Similar callsigns — climb clearance taken by the wrong aircraft

Clues from the past cases

I Converging traffic not detected
needed by CASE-02 (Conflict not detected in time) · priority 0.67

L Trainee on position
needed by CASE-07 (Incorrect readback not challenged) · priority 0.67

R No similar-callsign marking
needed by CASE-01 (Similar-callsign risk not controlled) · priority 0.00

O No callsign de-confliction with operators
needed by CASE-01 (Similar-callsign risk not controlled) · priority 0.00

R No criteria for combining at night
needed by CASE-02 (Sector-configuration risk not controlled) · priority 0.00

O Night roster at minimum staffing
needed by CASE-02 (Sector-configuration risk not controlled) · priority 0.00

R Readback confirmation not required
needed by CASE-07 (Readback risk not controlled) · priority 0.00

R Instructor not monitoring readback
needed by CASE-07 (Readback risk not controlled)

Expert system — AcciMaps and clues

ARAP

AirNav Risk Analysis Platform

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Ayu Assessor assessor

Investigation expert system (ORLIO)

Investigation expert system — OCC-ILL-01

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Occurrence category

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Advisory only: the system points to similar past occurrences and to evidence worth collecting. Investigators draw the conclusions; findings do not apportion blame or liability. Analysed 2026-09-29 08:01:48+00:00.

1 · Occurrence & search

2 · Evidence ratings (7)

3 · AcciMaps (fuzzy inference)

4 · Bayesian network & ranked clues

Mechanism: prior vs posterior · ...

Open as BBN study

Uncertainty H(C | evidence) = 2.2319 bits · evidence used: L-SIMILAR-CALLSIGN, L-COMBINED-SECTORS, L-NIGHT-FATIGUE, I-HEARBACK-MISSED, L-FREQ-CONGESTION, E-LOS · casebn-1.0.0

Ranked clues — evidence worth collecting next (value of information)

#	Factor	P(yes)	VOI bits	If yes / no	Rate
1	I Conflict not detected — monitoring lapse	40%	0.131	M5 37% / M2 55%	Not provided
2	L High workload / traffic peak	40%	0.093	M2 28% / M2 55%	Not provided
3	I Coordination not carried out	32%	0.064	M2 35% / M2 49%	Not provided
4	R Coordination procedure or LoA unclear	32%	0.064	M2 35% / M2 49%	Not provided
5	L Trainee controller on position	41%	0.054	M2 54% / M2 38%	Not provided
6	R OJT supervision / intervention criteria inadequate	41%	0.054	M2 54% / M2 38%	Not provided
7	I Pilot deviated from the cleared level	30%	0.043	M2 37% / M2 48%	Not provided
8	I Controller issued a conflicting level or clearance	30%	0.043	M2 37% / M2 48%	Not provided

Expert system — Bayesian network and ranked clues

Demo project **DEMO-06** is an illustrative loss of separation between two aircraft with similar callsigns on combined night sectors. The library has nine illustrative ATS cases and the reference example from Ng et al. (2022).

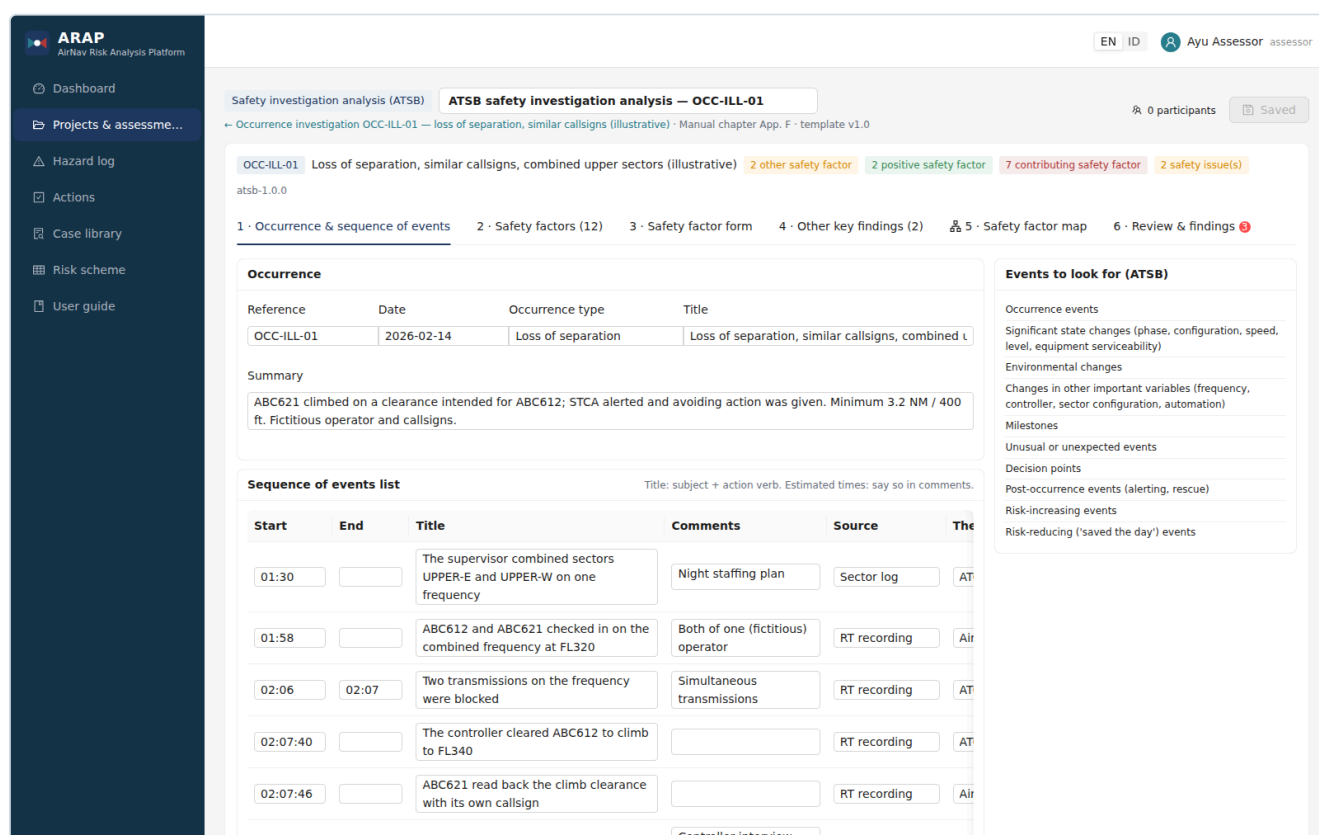
After the day-1 ratings:

- The network puts *readback/hearback error not detected* (M2) at 44% (prior 20%).
- CASE-01 gives M2 *Very likely*.
- The top clue is whether the conflict went **undetected while monitoring**, which best separates M2 from *conflict not detected in time* (M5).

Continue the investigation with SOAM/HFACS (§13.13). After the final report, add the occurrence to the case library.

13.17 Safety investigation analysis — the ATSB method (Manual Appendix F)

This study carries an occurrence investigation from the sequence of events to the findings, safety issues and safety action, following the ATSB *Safety Investigation Guidelines Manual — Analysis* and its tools manual. Add **Safety investigation analysis (ATSB)** to the investigation's assessment. NAVRAP re-checks the analysis automatically about half a second after each change.



Sequence of events

1 • Occurrence & sequence of events. Record the occurrence, then list the events. Each event needs a start time (estimate if necessary and say so in the comments) and a title written as a subject and an action verb. Theme groups events on the timeline (ATC, aircraft, crew). To put an event on the safety factors list, choose its type in the *Safety factor* column and click **Add**. The *Events to look for* list on the right is ATSB's prompt list.

2 • Safety factors. The master register: one row per potential safety factor with its type, codes, whether it is analysed further, existence, the **finding type** that NAVRAP derives, safety issue and risk level, and an *analysis complete* switch. Click an ID to open the factor.

ARAP

AirNav Risk Analysis Platform

Dashboard

Projects & assessments...

Hazard log

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EN ID

Ayu Assessor assessor

Safety investigation analysis (ATSB)

ATSB safety investigation analysis — OCC-ILL-01

0 participants

Saved

Occurrence investigation OCC-ILL-01 — loss of separation, similar callsigns (illustrative) · Manual chapter App. F · template v1.0

OCC-ILL-01 Loss of separation, similar callsigns, combined upper sectors (illustrative) 2 other safety factor 2 positive safety factor 7 contributing safety factor 2 safety issue(s)

atsb-1.0.0

1 · Occurrence & sequence of events 2 · Safety factors (12) 3 · Safety factor form 4 · Other key findings (2) 5 · Safety factor map 6 · Review & findings

+ Safety factor Other safety factor: 2 Not analysed further: 1 Positive safety factor: 2 Contributing safety factor: 7

ID	Title	Type	Codes	Further analysis	Existence	Finding type	Safety issue	Analysis complete
F1	ABC621 climbed from FL320 on a clearance intended for ABC612	Occurrence event		☒	yes	Contributing safety factor		☐
F2	The flight crew of ABC621 accepted and read back the climb clearance intended for ABC612	Individual action	I1.6	☒	yes	Contributing safety factor		☐
F3	The controller did not detect that the readback of the climb clearance came from ABC621	Individual action	I3.3	☒	yes	Contributing safety factor		☐
F4	Two aircraft of one operator with callsigns differing only in the order of the last two digits were on the same frequency	Local condition	L3.6	☒	yes	Contributing safety factor		☐
F5	Two transmissions were blocked on the combined frequency in the minute before the clearance	Local condition	L3.6	☒	yes	Contributing safety factor		☐
F6	The controller was in the last two hours of a night duty that began at 22:00	Local condition	L1.3	☒	yes	Other safety factor		☐
F7	Sectors were combined at 01:30 with traffic still above the daytime combining threshold	Local condition	L3.1	☒	yes	Other safety factor		☐
F8	The unit had no procedure or display aid to alert controllers to similar callsigns on the same frequency	Risk control	R3, R1.1	☒	yes	Contributing safety factor	Significant	☐

Safety factors list

3 · Safety factor form. For the selected factor:

- **Define:** a complete, neutral sentence (no "failed", "inadequate"; one factor per statement); the type; one or more ATSB codes from the taxonomy (coding guidance appears for codes that have it); the role and error type for individual actions, the functional area for conditions, preventive or recovery for risk controls, internal or external for organisational influences. Tick *Potential safety issue* (local conditions, risk controls and organisational influences only) and name the owner. Untick *Analyse further* only with a reason.
- **Existence and Influence:** evidence tables. Rate each item *supports*, *opposes*, *no effect* or *unsure*, and record its source, type, relevance, credibility concerns and whether it was expected but not seen. Then conclude *supported* or *not supported* with a probability expression (a finding needs *likely*, 66 % or more). For influence, first choose what the factor influenced: the occurrence or another safety factor. The questions on the right are ATSB's checklist for the test.
- **Importance** (only when existence passed and influence did not): is the factor still worth analysing? Justify the decision.
- **Explain:** the factors whose influence is on this one; **Add an explaining factor** starts a new one already linked.
- **Risk analysis** (safety issues): worst possible scenario, existing controls, worst credible scenario, then click the matrix. Choose **AirNav 5×5** (default) or **ATSB 6×6**. Both give *critical*, *significant* or *broadly acceptable* (not a safety issue). The small matrix records an alternative rating for sensitivity.
- **Safety action:** the issue status; each action with its kind (organisation's action, recommendation, advisory notice), organisation, date notified, status, classification and a communication log; **Track in Actions** creates an entry in the action list. Rate the residual risk, tick ALARP if justified and note the practicability considerations. NAVRAP shows whether further action is required and the six-monthly follow-up date.

Hazard log

Actions

Case library

Risk scheme

User guide

OCC-ILL-01 Loss of separation, similar callsigns, combined upper sectors (illustrative) 2 other safety factor 2 positive safety factor 7 contributing safety factor 2 safety issue(s)

atsb-1.0.0

1 · Occurrence & sequence of events
2 · Safety factors (12)
3 · Safety factor form
4 · Other key findings (2)
5 · Safety factor map
6 · Review & findings

F9 Similar-callsign de-confliction was not coordinated between the air navigation service provider and the operator

Organisational influence

Contributing safety factor

Significant safety issue

3 check(s)

- Similar-callsign de-confliction was not coordinated between the air navigation service provider and the operator: existence (66%+) and influence (66%+) together may be only about 44% — reconsider the contributing finding
- Similar-callsign de-confliction was not coordinated between the air navigation service provider and the operator: follow-up overdue since 2026-09-18 (every 6 months while above ALARP)
- Similar-callsign de-confliction was not coordinated between the air navigation service provider and the operator: consider reasonableness/practicability for this organisational finding

Define

Existence yes

Influence yes

Importance

Explain (0)

Risk analysis Significant

Safety action (1)

1. Worst possible scenario (ignoring controls)

Mid-air collision of two passenger aircraft.

4-6. Consequence x likelihood — click a cell

	V. large catastrophic A	Catastrophic B	Hazardous C	Major D	Minor E	Negligible F
Very frequent 6	6A	6B	6C	6D	6E	6F
Frequent 5	5A	5B	5C	5D	5E	5F
Occasional 4	4A	4B	4C	4D	4E	4F
Remote 3	3A	3B	3C	3D	3E	3F
Improbable 2	2A	2B	2C	2D	2E	2F
Extremely improbable 1	1A	1B	1C	1D	1E	1F

2B · Significant safety issue

Communicate; follow up through regular liaison; recommendation with the final report if not ALARP. Consequence: Catastrophic.

Sensitivity: an alternative rating if the team is uncertain

	V. large catastrophic A	Catastrophic B	Hazardous C	Major D	Minor E	Negligible F
Very frequent 6	6A	6B	6C	6D	6E	6F
Frequent 5	5A	5B	5C	5D	5E	5F
Occasional 4	4A	4B	4C	4D	4E	4F

2. Existing risk controls and their effectiveness

Control

Effectiveness

Controller readback/hearback

Vulnerable to similar callsigns

+ Control

3. Worst credible scenario (allow one more control failure, not several)

Clearance taken by the wrong aircraft with conflicting traffic; separation lost before STCA alerts.

Consequence justification

Catastrophic outcome credible for passenger operations if STCA and ACAS both fail to resolve.

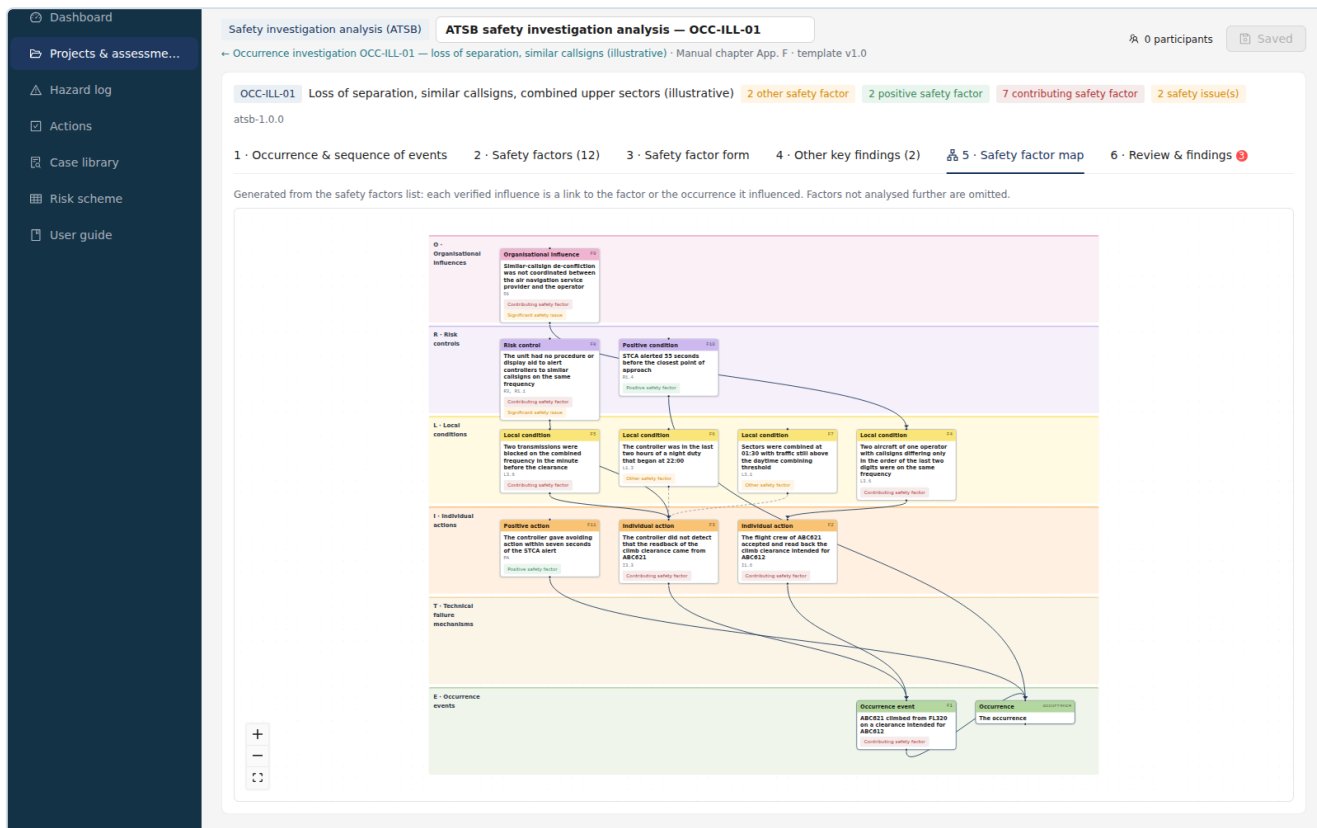
Likelihood justification (exposure × probability; compare with data)

Improbable: requires similar callsigns, a hearback error and failure of two independent controls.

Risk analysis on the ATSB 6×6

4 · Other key findings. Basic evidence tables for findings that are not safety factors.

5 · Safety factor map. Generated from the verified influences, in the ORLIO lanes.



Safety factor map

6 · Review & findings. The findings organised for the report; the safety issues with **Send to hazard log** (AirNav-rated issues keep their 5×5 rating; ATSB-rated issues carry the 6×6 rating in the text, to be rated on the AirNav scheme in the log); all checks (errors, warnings, notes) with links to the factor; the analysis review checklist and the stop rule.

The screenshot shows the ATSB safety investigation analysis interface. The sidebar on the left contains navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Case library, Risk scheme, and User guide. The main content area is titled 'Safety investigation analysis (ATSB) - ATSB safety investigation analysis - OCC-ILL-01'. It shows the investigation details for 'OCC-ILL-01: Loss of separation, similar callsigns, combined upper sectors (illustrative)'. The progress bar indicates the current step is '6 · Review & findings'. The findings are organized into sections: 'Contributing safety factors' (7 items), 'Other safety factors' (2 items), and 'Other key findings' (3 items). A 'Checks' section shows 0 errors, 3 warnings, and 2 notes. An 'Analysis review' section contains a checklist with items like 'Completeness: Test for sufficiency', 'Fairness: No finding appears to blame an individual', and 'Organise findings'.

Review and findings

How NAVRAP derives the finding type: existence + influence on the occurrence or a contributing factor → **contributing safety factor**; influence only on a non-contributing factor, or existence + importance → **other safety factor**; positive actions and conditions → **positive safety factor**; existence not shown → *not established*. The checks include arguments from ignorance (supported with no supporting item), conclusions below the standard of proof, two "just likely" tests that together fall below 50 %, evidence from one source, judgemental wording, the **test for sufficiency** (a contributing factor with no explaining factor) and **fairness** (an individual action with no identified reasons).

The ATSB 6×6 option. The Risk scheme page shows it beside the AirNav matrix. ATSB does not publish its cell colours as text, so NAVRAP uses the AirNav colours by the same pattern: critical = intolerable red; significant = amber (near critical) and yellow, like AirNav's two tolerable bands; broadly acceptable = green. Every catastrophic cell is at least significant.



13.18 Investigation report and the KNKT worked example (DEMO-07)

1. **Executive summary** — synopsis (what, when, where, who), consequences (injuries, damage, environmental impact) and the core finding. Unless you write your own, the core finding is generated from the ORLIO analysis: the contributing factors from the occurrence back to the organisational level, and the number of safety issues.
2. **Factual information** — the occurrence timeline (from the sequence of events list), personnel and assets (experience, equipment types, environmental configuration) and the immediate actions taken (emergency response, site preservation, stabilisation).
3. **ORLIO analysis framework** — 3.1 occurrence events (occurrence events and technical failure mechanisms), 3.2 individual actions, 3.3 local conditions, 3.4 risk controls and 3.5 organisational influences. Each layer shows its definition, your narrative for the layer, then each factor with its codes, finding, existence and influence results, key evidence, the factor it influenced and, for safety issues, the risk. For individual actions the report shows **why the action made sense to the person at the time** — enter this in the *Define* tab (field shown for individual and positive actions). It is written without assigning blame.
4. **Findings and contributing factors** — contributing factors in priority order (safety issues by risk level first, then from the organisational level down to the occurrence), each with its **chain** (e.g. F13 → F10 → F19 → F1 → the occurrence); then other factors that increased risk, other key findings, the safety issues and the factors considered but not included.
5. **Safety recommendations and corrective actions** — a table with the columns *ORLIO layer* · *Identified deficiency* · *Corrective action required* · *Action owner* · *Target date*. Each action is prefixed with its level in the **hierarchy of controls** (elimination, substitution, engineering,

administrative, PPE) and ordered by it. A blank target date prints as *[Date]*. A contributing factor with no action gets a flagged row (*[No corrective action recorded]*), and NAVRAP also lists it as a note in the checks.

6. **Appendices** — A: photographs and diagrams you upload (PNG/JPEG up to 3 MB, with captions); B: interview summaries and witness accounts; C: the **ORLIO factor map**, drawn automatically (solid borders = contributing, dashed = other; solid arrows = established influence; SI badges = safety issues with their risk index).

Corrective actions are entered in the factor's **Safety / corrective action** tab, now available for every factor (the issue status and residual-risk evaluation appear only for safety issues). Each action has a hierarchy level, a target date and a reference (for example a recommendation number).

Click **Preview** to see the report beside the form, or **Word (.docx)** / **PDF** to download it. The current content is used even if it is not yet saved; downloads are recorded in the audit trail.

ARAP
AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

Safety investigation analysis (ATSB) **ORLIO analysis (ATSB method) — KNKT.24.10.22.04** 0 participants Saved

← ORLIO safety investigation — PK-GMP / lavatory service truck collision, 16 October 2024 · Manual chapter App. F · template v1.0

KNKT.24.10.22.04 Boeing 737-800 PK-GMP struck by an unattended lavatory service truck at parking stand W29, Kualanamu (WIMM) 6 other safety factor 11 contributing safety factor

7 safety issue(s) atsb-1.1.0

1 · e of events 2 · Safety factors (19) 3 · Safety factor form 4 · Other key findings (3) 5 · Safety factor map 6 · Review & findings 7 · Investigation report ...

Investigation report
Sections 3-5 and Appendix C are generated from the analysis (safety factors, tests, safety issues and corrective actions). Enter here the factual information the analysis does not hold. Structure: 1 Executive summary · 2 Factual information · 3 ORLIO analysis framework · 4 Findings & contributing factors · 5 Safety recommendations & corrective actions · 6 Appendices.

Report and synopsis (1)

Report number	Status	Prepared by
ARAP-ORLIO-KNKT.24.	Example (not an ...)	ARAP worked example — ORLIO a

Where Time Who (organisations)

Kualanamu International Airp about 2126 PT Garuda Indonesia (aircraft)

Consequences — injuries, damage, environmental impact

No injuries to the 109 persons on board (2 pilots, 5 flight attendants, 102 passengers) or to ground staff. The tail section of PK-GMP and the lavatory service truck were damaged. No environmental impact was reported.

Core finding (optional — leave empty to use the summary generated from the ORLIO analysis)

Source / basis of the report

KNKT final report KNKT.24.10.22.04 (Komite Nasional Keselamatan Transporta

Generate

Preview Word (.docx) PDF Close preview

Uses the current (also unsaved) content. Exports are recorded in the audit trail.

Safety Investigation Report
Boeing 737-800 PK-GMP struck by an unattended lavatory service truck at parking stand W29, Kualanamu (WIMM)
ARAP-ORLIO-KNKT.24.10.22.04 · DEMO-07 — Case study: KNKT.24.10.22.04 — B737-800 PK-GMP struck by a lavatory service truck, Kualanamu · Example (not an official report) · ARAP worked example — ORLIO analysis using the ATSB method, based on the KNKT final report · generated 2026-10-01

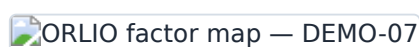
1. Executive summary

1.1 Synopsis

What	Boeing 737-800 PK-GMP struck by an unattended lavatory service truck at parking stand W29, Kualanamu (WIMM)
Occurrence type	Ground collision — GSE with aircraft (serious incident)
When	2024-10-16 about 2126 LT (UTC+7)
Where	Kualanamu International Airport (WIMM), Deli Serdang, North Sumatra, Indonesia
Who	PT Garuda Indonesia (aircraft); PT Garuda Angkasa (ground handling); PT Angkasa Pura Aviast (airport)
Reference	KNKT.24.10.22.04

Report tab with preview

DEMO-07 — worked example from KNKT.24.10.22.04. The demo project contains a complete stand-alone ORLIO analysis, using the ATSB method, of the serious incident at Kualanamu on 16 October 2024, in which an unattended lavatory service truck (LST) rolled and struck the tail of Boeing 737-800 PK-GMP at parking stand W29. Facts, findings, safety actions taken and the seven safety recommendations (04-G-2024-22.01 to .03, 04-B-2024-22.04, 04-O-2024-22.05, 04-R-2024-22.06 and .07) are taken from the KNKT final report. The evidence ratings, probability expressions, risk ratings, hierarchy levels and the actions marked *proposed* (NAVRAP example) are NAVRAP's illustration. Two judgements differ from KNKT and are explained in the factors: the vibration/engine-suction trigger and the operator's fatigue are *other* factors, because the evidence did not reach *likely* under the ATSB standard of proof. A generated report is in docs/examples/.



ORLIO factor map — DEMO-07

13.19 Worked example: surveillance radar performance degradation (DEMO-08)

DEMO-08 follows one technical problem through the whole life of a change and uses every method that is relevant to each phase and each type of hazard (Manual Appendix G). The fictitious Bukit Sari MSSR, which feeds ACC Sector E3, degrades gradually: lower probability of detection in the north-east quadrant, azimuth bias, reflected false targets and Mode C garbling. Its monitoring alarms only on hard failures, so no alarm is raised. A loss of separation follows; the radar is refurbished in night outage windows and returned to service.

The screenshot shows the NAVRAP AirNav Risk Analysis Platform interface. The left sidebar contains navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Case library, Risk scheme, and User guide. The main content area displays the project details for DEMO-08 — Surveillance radar performance degradation — Bukit Sari MSSR (worked example). The project is active, sponsored by the Directorate of Operations and Directorate of CNS, and involves units ACC Sector E3 (FL100-FL245); APP-5; CNS engineering (radar site). The description details the radar's degradation and the resulting loss of separation. Below the project details is a table of assessments.

Assessment	Version	Studies	Status	Created by
P1 · Concept — degraded radar and the refurbishment decision	1	4	draft	assessor
P2 · Design — degraded-mode operation and the refurbished surveillance chain	1	12	draft	assessor
P3 · Implementation and transition — radar refurbishment works and return to service	1	4	draft	assessor
P4 · Operations — living with residual degradation	1	2	draft	assessor
P5 · Occurrence investigation — OCC-RDR-01 loss of separation	1	3	draft	assessor

DEMO-08 project

The project has **six assessments, one per phase**. Each assessment's scope lists its studies with the hazard types they cover: TEC technical, HUM human, PRO procedural, ORG organisational, ENV environmental, SEC security, OHS occupational.

Phase	Studies
P1 Concept	HAZID, FHA, bowtie, expert judgement (Cooke and Delphi)
P2 Design of the degraded mode and the refurbished chain	FMEA, FTA, common cause analysis, RBD and Markov, STPA, HAZOP, security risk, CRM, LOPA, ETA, HTA, HRA
P3 Implementation and transition	JHA of the tower works, SWIFT of the outage windows and return to service, real-time simulation, ATSEP fatigue
P4 Operations	FRAM of work-as-done, BBN for diagnosing the cause from the symptoms
P5 Occurrence investigation	ERC/RAT, ATSB analysis of OCC-RDR-01 (with the investigation report), SOAM/HFACS/Tripod
P6 Monitoring and argument	SPI register, GSN safety argument linking all 26 studies

 Methods by phase and hazard type
Methods by phase and hazard type

Key results to look at:

- **FTA:** about 9×10^{-6} per flight hour in the degraded state, and ten times less with performance thresholds.
- **Common cause analysis and LOPA:** STCA is not credited as an independent layer, because it uses the same tracks as the controller.
- **Markov:** finding the degradation within a week instead of two months raises in-specification availability from 0.82 to 0.96.
- **CRM:** the parallel routes 10 NM apart need 11.7 NM without radar monitoring.
- **Fatigue:** eight-hour nights are better than twelve-hour nights for the works crew.

The hazard log holds RDR-01 to RDR-12, each with its type and phase in the *context* field, its source study and row, its controls and its actions (RDR-A01 to A08). Wildlife strike risk, the organisational function map and the expert system are not used; the project description says why.

The screenshot displays the NAVRAP AirNav Risk Analysis Platform interface. The left sidebar contains navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Case library, Risk scheme, and User guide. The main content area is titled 'P2 · Design — degraded-mode operation and the refurbished surveillance chain' (draft v1). Below the title, there's a breadcrumb trail: DEMO-08 — Surveillance radar performance degradation — Bukit Sari MSSR (worked example). A 'Submit for review' button and a 'Report (.docx)' download link are visible. The 'Scope, environment and assumptions' section includes a detailed description of the scope, environment (ACC Sector E3, APP-S, Bukit Sari MSSR), and assumptions (Hazard types covered: Environmental, Human performance, Organisational, Procedural / operational, Security, Technical). The 'Method studies (12)' section lists suggested studies for a system change, including HAZID, FHA, FMEA / FMECA, FTA, Common cause analysis, RBD and Markov analysis, Bowtie analysis, Security risk assessment, and Safety case (GSN). A table lists the studies with columns for Method, Study, Status, and Updated. The 'Risk acceptance' section shows the worst current/residual region as 'Tolerable (upper)' and the acceptance authority as 'Accountable Executive'. The 'Actions (4)' section lists four actions (RDR-A02, RDR-A03, RDR-A04, RDR-A08) with their descriptions and due dates. The 'Review and approval history' section shows 'Not yet submitted.'

Method	Study	Status	Updated
FMEA / FMECA	FMEA — radar sensor to controller display chain	open	2026-10-02 09:59
FTA	FTA — undetected erroneous position displayed in Sector E3	open	2026-10-02 09:59
Common cause analysis	Common cause analysis — radar, ADS-B, STCA and the controller's cross-check	open	2026-10-02 09:59
RBD and Markov analysis	RBD and Markov — surveillance availability and the cost of late detection	open	2026-10-02 09:59
STPA	STPA — declaring and applying degraded		

Design phase

14. The hazard log

ARAP
AirNav Risk Analysis Platform

Dashboard

Projects & assessments...

Hazard log

Actions

Risk scheme

User guide

EN ID Ayu Assessor assessor

+ New hazard CSV

Hazard log

Single register of hazards from every method study, with current and residual risk, controls and owners (Manual §18.2).

Search

Risk region

6 of 6

Ref	Hazard	Unit	System	Owner	Initial	Residual	Controls	Review	Status
HZ-0001	Unauthorised entry onto active runway	Soekarno-Hatta TWR	Aerodrome	TWR unit chief	3A	2A	2	2026-10-18	open
HZ-0002	Level bust leading to potential mid-air collision	Jakarta ACC	ATM	ACC unit chief	3A	1A	2	2027-01-26	open
HZ-0003	Total loss of surveillance picture at ACC sector	Jakarta ACC	Surveillance	ACC unit chief	3B	3B	1	2026-11-12	open
HZ-0004	ATCO fatigue at end of night shift	Jakarta ACC	Roster	ACC unit chief	4C	3C	1	2026-09-23	open
HZ-0005	New site outside radio line of sight of low-level sector	Jakarta APP	VHF	ACC unit chief	3B	—	0	2026-11-27	open
HZ-0006	Loss of surveillance display at cut-over	Jakarta APP	Surveillance	Project manager	3C	2C	1	2026-12-27	open

< 1 >

Hazard log

The hazard log is the single register of hazards from all studies (Manual §32.2). Search, filter by risk region, sort, and export to **CSV**. Overdue review dates are shown in red.

Click a hazard to open it:

ARAP

AirNav Risk Analysis Platform

Dashboard

Projects & assessments...

Hazard log

Actions

Risk scheme

User guide

Hazard log

Single register of hazards from every method study, with current

Search

Risk region

Ref	Hazard	Unit
HZ-0001	Unauthorised entry onto active runway	Soeka
HZ-0002	Level bust leading to potential mid-air collision	Jakarta
HZ-0003	Total loss of surveillance picture at ACC sector	Jakarta
HZ-0004	ATCO fatigue at end of night shift	Jakarta
HZ-0005	New site outside radio line of sight of low-level sector	Jakarta
HZ-0006	Loss of surveillance display at cut-over	Jakarta

×

HZ-0001 — Unauthorised entry onto active runway

Save

Details and risk

Controls (2)

Links

History

* Hazard

Unauthorised entry onto active runway

Causes / threats

Consequences

Unit

System

Owner

Soekarno-Hatta TWR

Aerodrome

TWR unit chief

Status

Review date

open

2026-10-18

Initial / current risk

Residual risk

clear

	Catastrophic	Hazardous	Major	Minor	Negligible
	A	B	C	D	E
Frequent 5	5A	5B	5C	5D	5E
Occasional 4	4A	4B	4C	4D	4E
Remote 3	3A	3B	3C	3D	3E
Improbable 2	2A	2B	2C	2D	2E
Extremely improbable 1	1A	1B	1C	1D	1E

	Catastrophic	Hazardous	Major	Minor	Negligible
	A	B	C	D	E
Frequent 5	5A	5B	5C	5D	5E
Occasional 4	4A	4B	4C	4D	4E
Remote 3	3A	3B	3C	3D	3E
Improbable 2	2A	2B	2C	2D	2E
Extremely improbable 1	1A	1B	1C	1D	1E

Rationale and evidence for the ratings (required for residual risk)

Occurrence data 2023–2025; barrier audit results

Description / context

Hazard detail

- **Details and risk** — description, causes, consequences, unit, system, owner, status and review date. Rate **initial/current** and **residual** risk by clicking a cell in each matrix. A residual rating requires a **rationale** (the basis of the judgement).
- When you create a new hazard, NAVRAP checks for **possible duplicates** as you leave the title field.
- **Controls** — each with side (prevention/recovery), type, status, effectiveness, owner and a safety-critical flag. Only *existing-verified* controls may be credited in the current risk.
- **Links** — the source study, the assessment and the related actions.
- **History** — every change from the audit trail.

15. Actions

ARAP
AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

Actions
Risk-reduction actions with owners and due dates. Overdue actions are escalated on the dashboard (COM-12).

+ New action

Ref	Action	Hazard	Owner	Due	Status
ACT-0003	Collect KSS survey for two roster cycles	HZ-0004	FRMS specialist	2026-09-18	open
ACT-0004	Audit stop-bar U/S procedure compliance	HZ-0001	Aerodrome ops	2026-09-25	In progress
ACT-0005	Radio coverage study for new site	HZ-0005	CNS engineering	2026-10-28	open
ACT-0001	Install diverse, physically separate surveillance data path	HZ-0003	CNS engineering	2026-12-27	open
ACT-0002	Implement SFL mismatch alert in ATM system	HZ-0002	ATM systems	2027-02-25	open

< 1 >

Actions

Actions have an owner, a due date, a status (open, in progress, closed) and an optional link to a hazard. Overdue actions are flagged here and on the dashboard. **Closing an action requires closure evidence** (e.g. a report number). A Viewer can update an action when the action's owner field is their username.

16. Review, approval and locking

Status	Meaning	Next step and who does it
Draft	Being prepared; editable.	Submit for review — assessor
In review	Reviewer checking.	Endorse or Return to assessor (comment required) — reviewer
Endorsed	Reviewed; locked.	Accept residual risk — authority for the worst residual region; or Reject (comment required)
Accepted	Residual risk accepted; locked.	Close — reviewer or authority
Rejected	Not accepted.	Revise and Submit again
Closed / Superseded	Finished, or replaced by a newer version.	—

Rules NAVRAP enforces

- An assessment with an **intolerable** residual risk cannot be accepted.
- An assessment with **no hazards** cannot be accepted.
- The person accepting must hold authority for the **worst** residual risk region in the assessment (e.g. *Tolerable (upper)* needs the Accountable Executive).
- Endorsed, accepted and closed assessments are **locked**: studies and hazards become read-only. To change anything, use **New version**. This copies the studies into a new draft (version n+1) and marks the old version *superseded*.

The screenshot shows the ARAP interface for a 'Transition safety assessment' (version 1) which is in an 'accepted' state. The interface includes a sidebar with navigation options: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme, and User guide. The main content area displays the assessment details, including the scope, environment, and assumptions. It also shows a table of method studies and a table of hazards. The risk acceptance section indicates the worst current/residual region is 'Tolerable (lower)' and the acceptance authority is 'Director of Operations / Engineering'. The actions section shows 'None'. The review and approval history shows an 'endorse by reviewer' and an 'accept by director'.

Accepted assessment

Every transition is recorded in the **review and approval history** and the audit trail.

17. Reports and exports

What	Where	Format
Safety Assessment Report (Manual §32.3)	Assessment page → Report (.docx)	Word
Hazard log	Hazard log → CSV	CSV (opens in Excel)
Bowtie diagram	Bowtie editor → PNG / SVG	Image
Investigation report (ATSB-method study, §13.18)	Study → 7 • Investigation report → Word / PDF	Word, PDF

The report includes scope, methods used, hazards and their risk ratings, study results (fault tree cut sets, LOPA verdict, fatigue indicators, Bayesian network probabilities, and for the extended methods the event-tree outcomes, collision risk, HEPs, availability, simulation statistics, expert weights, occurrence classifications, the GSN argument and the investigation findings), actions, the approval history, and the versions of the risk scheme and calculation engines used.

18. Administration

18.1 Risk scheme

Risk scheme shows the matrix, the tolerability regions with their required action and acceptance authority, and the likelihood levels with their quantitative bands.

The screenshot displays the ARAP (AirNav Risk Analysis Platform) interface. The left sidebar contains navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme (selected), and User guide. The main content area is titled 'Risk scheme' and includes a subtitle: 'Active version 1. Changes create a new version and never alter locked assessments (ADM-01).'.

Risk matrix

The risk matrix is a 5x5 grid. The columns are labeled 'Catastrophic', 'Major', 'Minor', and 'Negligible'. The rows are labeled 'Frequent', 'Occasional', 'Remote', 'Improbable', and 'Extremely improbable'. The cells are color-coded: red for 'Intolerable', orange for 'Tolerable (upper)', yellow for 'Tolerable (lower)', and green for 'Acceptable'. A tooltip for cell 5B indicates 'Intolerable'.

Tolerability and acceptance authority

Region	Cells	Required action	Authority
Intolerable	3A, 4A, 4B, 5A, 5B, 5C	Do not introduce or continue; redesign or add controls.	Cannot be accepted
Tolerable (upper)	2A, 3B, 4C, 5D	Additional controls; ALARP justification; monitoring plan.	Accountable Executive
Tolerable (lower)	1A, 2B, 2C, 3C, 3D, 4D, 4E, 5E	Mitigate where reasonably practicable; document ALARP.	Director of Operations / Engineering
Acceptable	1B, 1C, 1D, 1E, 2D, 2E, 3E	Accept; monitor.	Unit / branch manager

Likelihood levels

Level	Name	Nama	Band (per operating hour)
5	Frequent	Sering	1e-3 and above
4	Occasional	Kadang-kadang	1e-5 to 1e-3
3	Remote	Jarang	1e-7 to 1e-5
2	Improbable	Tidak mungkin	1e-9 to 1e-7
1	Extremely improbable	Sangat tidak mungkin	< 1e-9

Risk scheme

Administrators can reassign cells: choose a region under **Paint region**, click the cells, then **Save as new version**. NAVRAP checks that every cell belongs to exactly one region. Existing assessments keep the version they were created with.

18.2 Users

Users (administrators only) lists accounts. Click one to edit, or **New user** to add one. Set the role, unit and — for authorities — the risk regions they may accept. Deactivate rather than delete leavers.

18.3 Audit trail

Audit trail (reviewers and administrators) lists every create, change, approval and export with the user, time and changed fields. Filter by entity.

19. Troubleshooting and FAQ

I can't edit a study or hazard. Either your role is Viewer or Authority, or the assessment is endorsed/accepted/closed. Use **New version** on the assessment page (assessor or reviewer).

"A rationale is required for every risk judgement". Fill in *Rationale and evidence* in the hazard before saving a residual rating.

"Residual risk region ... requires ...". You do not hold authority for the worst residual risk in the assessment. Ask the named authority to accept it.

The worksheet did not keep my edit. Finish the edit (Enter or click another cell), then press **Save** in the header.

My session ended. Sessions last 8 hours by default; sign in again.

Numbers differ slightly from the Manual. The engines reproduce the Manual's worked examples within the tolerances in SRS §9. Rounding in the Manual text is to 2–3 significant figures.

20. What version 0.9 does not do yet

Version 0.9 covers the core workflow and all twenty-six methods. The following SRS items are planned for later releases:

- Single sign-on through Keycloak / SAML / OIDC with MFA. Version 0.2 uses local accounts with JWT tokens.
- Real-time co-editing of a diagram by several users, and the offline workshop pack.
- Imports and exports in BowTieXP XML, Open-PSA MEF, XMLBIF, GeNIe and FRAM Model Visualiser formats.
- File attachments (MinIO), e-mail notifications and reminders.
- Integrations with the occurrence reporting, rostering and maintenance systems.
- CAST (STPA for accident analysis), noisy-OR helpers for Bayesian networks, Monte Carlo uncertainty, and the full “braked” sleep-recovery form of the fatigue model.
- An Apache Superset / Metabase reporting database.
- Linking SPI-register rows to bowtie barriers so barrier health shows the SPI status (SRS SPI-03).
- Deleting a single assessment (only whole projects can be deleted), and a retention period after which archived projects are purged.
- For the investigation expert system: meaning-based search with text embeddings (pgvector), network structures other than mechanism → factors (e.g. noisy-OR between factors), statistical validation reports against held-back cases, and AI drafting tested only against a simulated service in the automated tests.
- For the ATSB analysis: rail and marine taxonomies, editing the report's section wording, and administrator editing of the ATSB 6×6 cell assignment (the matrix is fixed in this version).
- Saving an existing assessment as a new template, and administrator editing of templates (SRS TPL-05).
- For the wildlife module: import of strike records from the occurrence system and in ICAO IBIS format.
- For the extended methods: import of occurrences from the reporting system into the ERC/RAT module, CRM sensitivity cases, administrator editing of the CARA/HEART libraries, and restricted (need-to-know) access to security studies (SRS SRA-04).